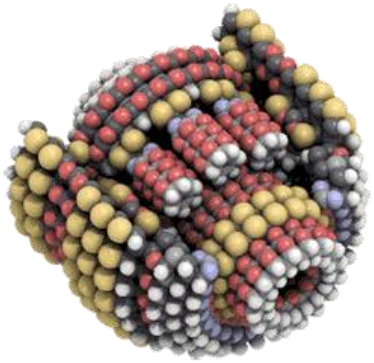
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MATE6301 – Nanomaterials



Week 6: Application of Nano-magnetism – Nano-machineries

Assoc. Prof. M Shahriar Hossain
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Outline

The particle size effect on magnetic properties and determination of critical size.

Magnetic properties characterization techniques.

Nano-machineries.

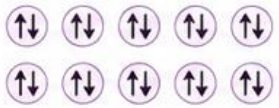
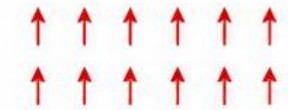
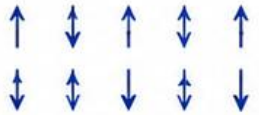
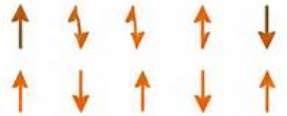
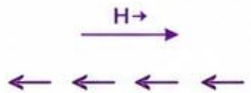
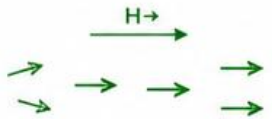
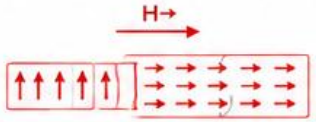
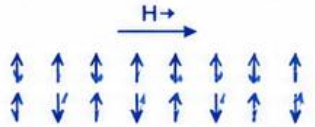
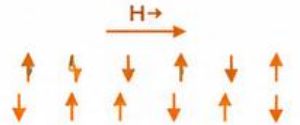
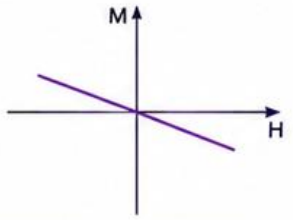
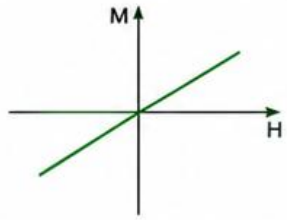
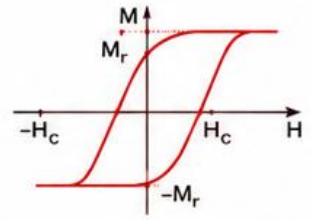
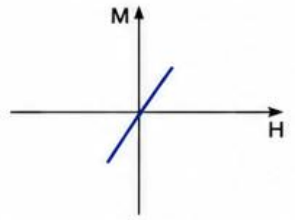
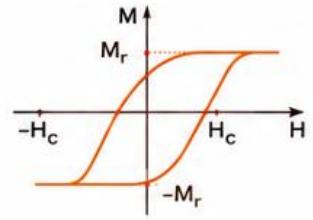
Case Study 4. Magnetic hyperthermia for cancer treatment.

Case Study 5. Magnetically triggered drug delivery system by using combined AC and DC magnetic field.

Lab prac :
Magnetic nanoparticles characterisation for biosensors.

Five Basic Types of Magnetism

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	DIAMAGNETISM	PARAMAGNETISM	FERROMAGNETISM	ANTIFERROMAGNETISM	FERRIMAGNETISM
Atomic / Spin Arrangement (H = 0)	No permanent magnetic moment (all electrons paired) 	Permanent moments randomly oriented 	Spins align parallel (spontaneous magnetization) 	Neighboring spins align antiparallel and cancel 	Neighboring spins antiparallel but unequal → net magnetization 
Response to Applied Field (H →)	Induced moment opposite to applied field 	Moments tend to align with the field 	Domains grow and align with the field 	Slight canting of spins; small net moment 	Unequal moments lead to net alignment 
M–H Curve (at 300 K)					
Key Comments	• No permanent dipole moment • Induced moment opposite to H • Weak repulsion from magnets • Present in all materials	• Permanent moments exist • Random due to thermal motion • Weak attraction to magnets • No hysteresis	• Strong parallel exchange interaction • Spontaneous magnetization below Curie temperature (T_C) • Hysteresis with large M_r and H_c	• Strong antiparallel exchange • Net moment ≈ 0 (if equal) • Below Néel temperature (T_N) • No hysteresis (or very small)	• Antiparallel but unequal moments • Nonzero net magnetization • Hysteresis similar to ferromagnets • Below Curie temperature (T_C)
Examples	Bi, Cu, Ag, Au, Water, Graphite	Al, Pt, Mg, Cr, O_2, rare earth salts	Fe, Co, Ni, Gd, Dy, Sm	MnO, NiO, FeO, Cr_2O_3, MnF_2	Fe_3O_4 (Magnetite), $CoFe_2O_4$, $NiFe_2O_4$
Magnetic Susceptibility (χ)	$\chi < 0$ (very small, ~10^{-6} to 10^{-5})	$\chi > 0$ (small, ~10^{-6} to 10^{-3})	$\chi >> 1$ (very large, nonlinear)	$\chi \approx 0$ (very small) (positive above T_N, negative below T_N)	$\chi >> 1$ (large, nonlinear)

Magnetism arises from electron spin and orbital motion and their exchange interactions. Temperature can destroy magnetic order: Ferromagnetism and ferrimagnetism disappear above Curie temperature (T_C); Antiferromagnetism disappears above Néel temperature (T_N).

Ferromagnetism to superparamagnetism (transition to nanomagnetism)

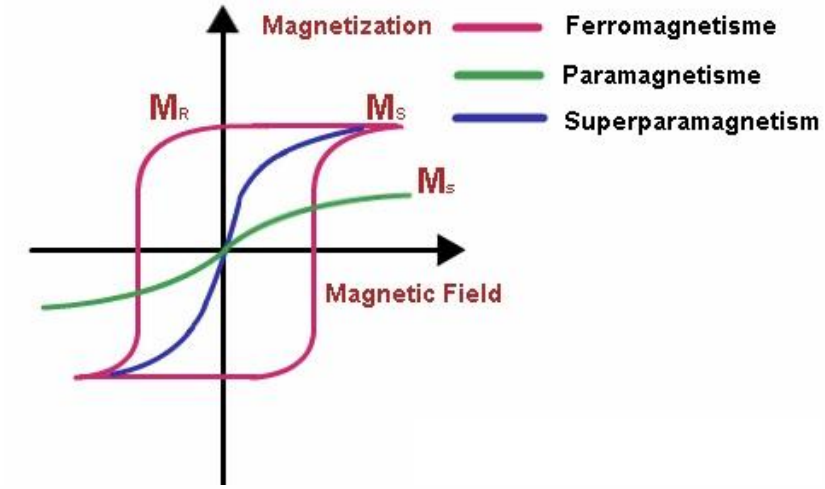
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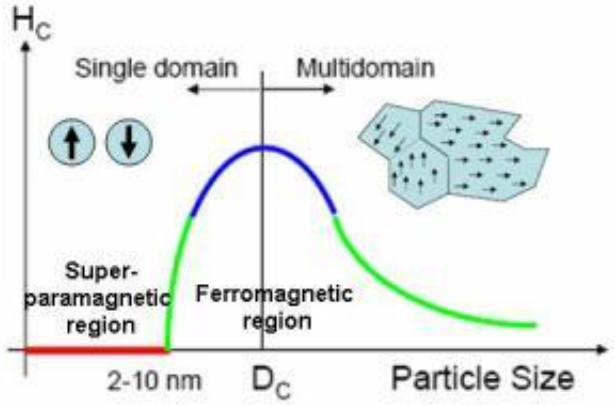
What is 'Super' about Superparamagnetism
Very fine ferromagnetic particles have very short relaxation times, even at room temperature, and behave superparamagnetic.

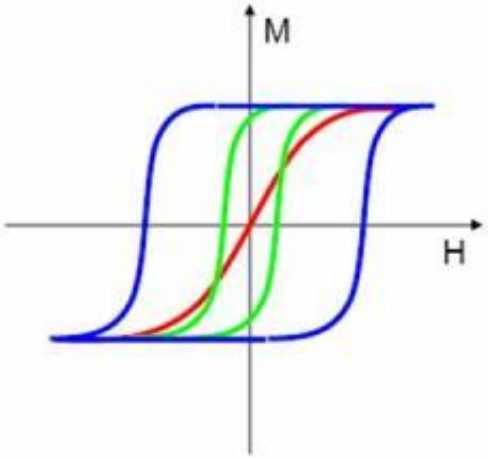
Superparmagnetic materials have:

- A high saturation magnetiszation (Ms)
- Zero coercivity (Hc) and
- Zero remanence (Mr),
- Single domain

Making it to be distinguished from ferromagnetism and paramagnetism - Superparamagnetism is a size effect of ferromagnetism.







The influence of magnetic particle size on magnetic properties. The coercivity changes with the particle size.

https://www.science20.com/mei/blog/blocking_temperatu

https://mappingignorance.org/2014/03/05/superparamagnetic-nanoparticles-and-the-separation-problem/

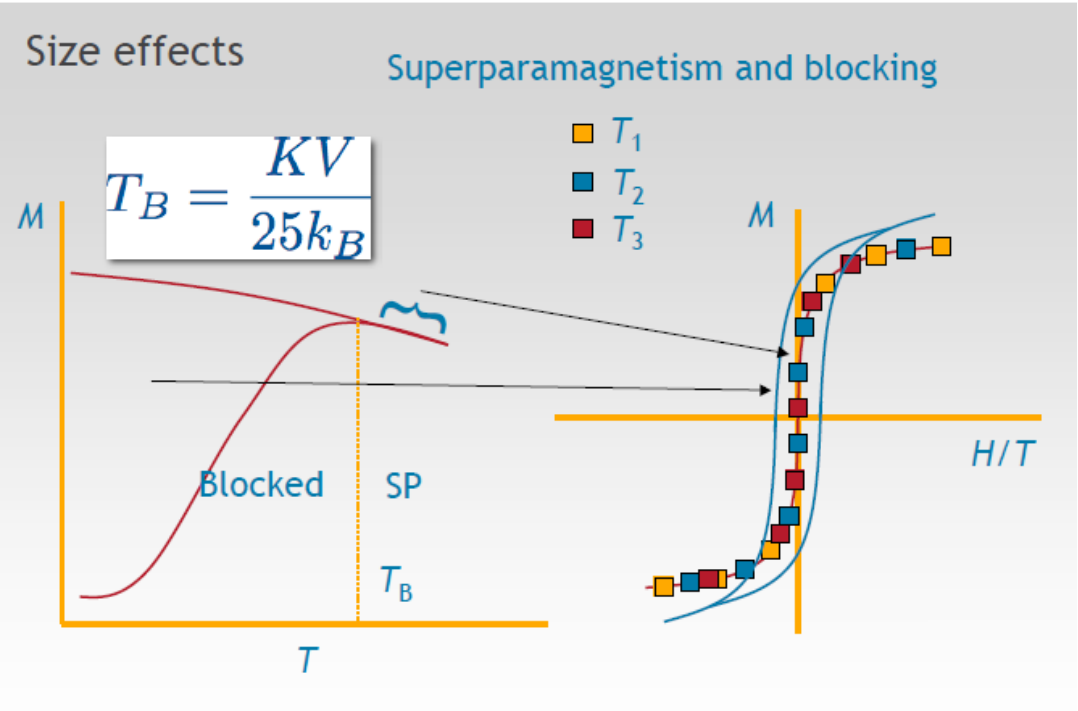
Ferromagnetism to superparamagnetism (transition to nanomagnetism)

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Some important features about Superparamagnetism

- Their behavior is paramagnetic (lower Ms) but their magnetization values are typical of ferromagnetic substances (higher Ms and zero Hc and Mr).

- The individual particles have normal ferromagnetic movements but very short relaxation times, enabling them to rapidly follow directional changes in an applied field.



- Magnetization curves measured at different temperatures superimpose when magnetization (*M*) is plotted as a function of field (*H*)/temperature (*T*).

- This also demonstrates that superparamagnetism can be destroyed by cooling.

- The temperature at which this occurs is called the blocking temperature (T_B) and is dependent linearly on volume and the magnitude of the crystal field anisotropy.

K is the effective magnetic anisotropy constant, V is the magnetic particle volume, kB is Boltzmann's constant, and TB is the **blocking temperature**.

Problem solving

You synthesize two batches of **F**e_3O_4 nanoparticles:
Batch A: diameter DA=8 nm
Batch B: diameter DB=20 nm

Assume spherical particles, room temperature T=300K, Cu Kα XRD confirms single-crystal particles, and the effective anisotropy energy density is $K=1.0 \times 10^5$ J m^{-3}
Use the "blocking criterion" based on the Néel–Arrhenius law with a typical measurement window ($\ln(\tau m/\tau 0)\approx 25$), i.e.
Tb = KV/25kb
KV=25kbTb

Energy barrier, Eb = KV
Thermal fluctuation energy, Et = 25kbTb
Blocked/ferromagnetic if $EB=KV \gtrsim 25$ kBT, superparamagnetic if $EB \ll 25$ kBT.

Tasks
1. Compute the particle volumes VA and VB.
2. Compute the energy barriers EB,A and EB,B.
3. Compare each to 25kBT and decide whether each batch is superparamagnetic or blocked at 300 K.

Given Constants:
$kB=1.380649\times10^{-23}$ J/K (Boltzmann constant)
$25kBT=25\times kB\times 300\approx 1.035\times10^{-19}$ J
Sphere volume $V=4/3\pi r^3$

Solution

Thermal energy, $25k_bT=25\times kB\times 300\approx 1.035\times 10^{-19}$ J
Energy Barrier, $E_b = KV$

1) Volumes
Batch A:
$D_A=8$ nm$\Rightarrow rA=4\times 10^{-9}$ m
$V_A=4/3\pi(4\times 10^{-9})^3=4.18879\times 64\times 10^{-27}\approx 2.681\times 10^{-25}$ m3

Batch B:
$D_B=20$ nm$\Rightarrow rB= 20/2$ nm $= 1.0\times 10^{-8}$ m
$V_B=4/3\pi(1.0\times 10^{-8})^3=4.18879\times 10^{-24}$ m3

2) Energy barrier, $E_b=KV$
$E_b,A=1.0\times 10^5\times 2.681\times 10^{-25}\approx 2.681\times 10^{-20}$ J
$E_b,B=1.0\times 10^5\times 4.18879\times 10^{-24}\approx 4.189\times 10^{-19}$ J

3) Compare to thermal energy, $25k_bT\approx 1.035\times 10^{-19}$ J
Batch A (8 nm): $E_b,A\approx 2.681\times 10^{-20}$ J
thermal energy, $25k_bT > E_b,A \rightarrow$ superparamagnetic at 300 K.

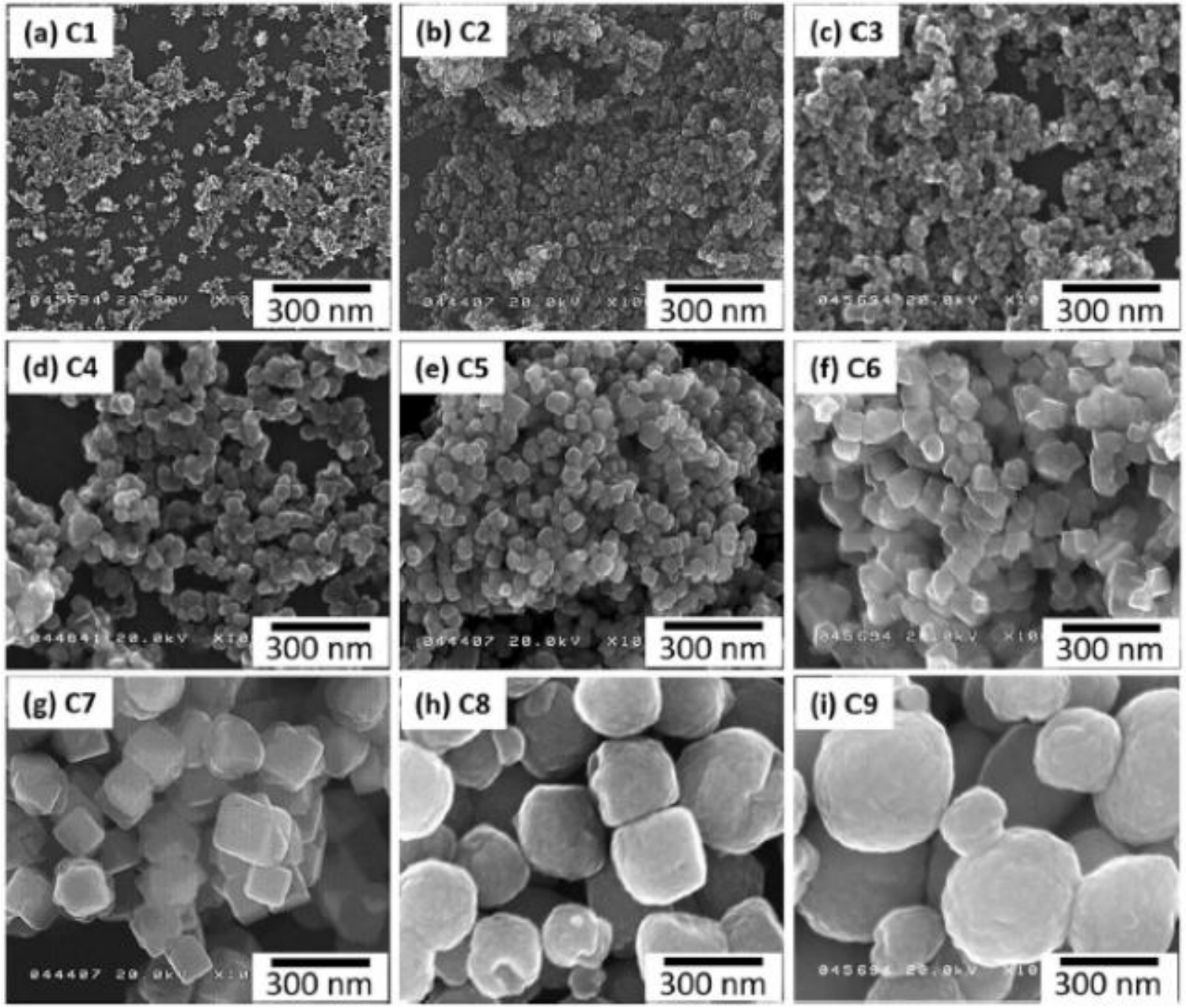
Batch B (20 nm): E_b, $B\approx 4.189\times 10^{-19}$ J
$25kBT< E_b$, $B \rightarrow$ blocked (ferromagnetic-like) at 300 K.
8 nm Fe_3O_4: superparamagnetic at room temperature.
20 nm Fe_3O_4: blocked at room temperature ferromagnetic (lost superparamagnetism).

Case study 3:

- **Highly crystalline single-domain 10-300nm magnetite Fe_3O_4 nanoparticles have been synthesized**
- **Large-scale liquid precipitation method was used**
- **Well-controlled crystal growth (by precise control of Fe^{2+} concentration, PH and temperature)**
- **Cubed and spherical shaped (By 2-stage and 1 stage oxidation reaction, respectively)**
- **Critical size was determined in terms of transforming from single and multi-domain structure**
- **Domain structures were determined by TEM using the lattice fringes**
- **Compared with cube-like NPs, sphere-like NPs have lower magnetic coercivity and remanence values, which is caused by the different orientations of their polycrystalline structure**

https://www.nature.com/articles/s41598-017-09897-5

Case study 3:



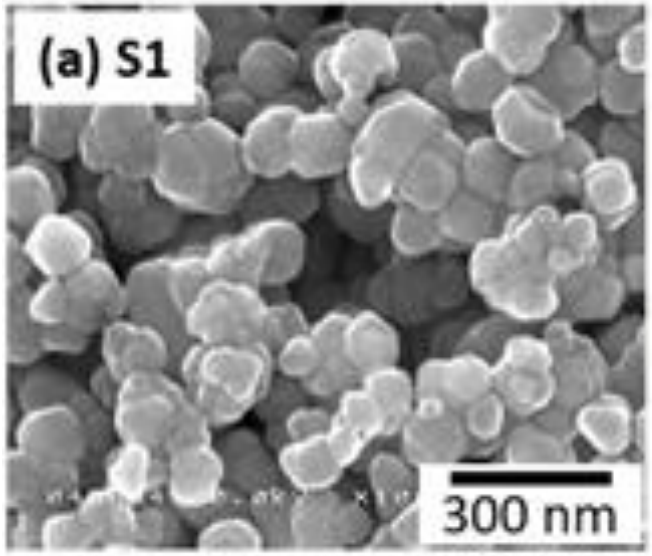
- Highly crystalline single-domain 10-300nm magnetite Fe3O4 nanoparticles have been synthesized using large-scale liquid precipitation method

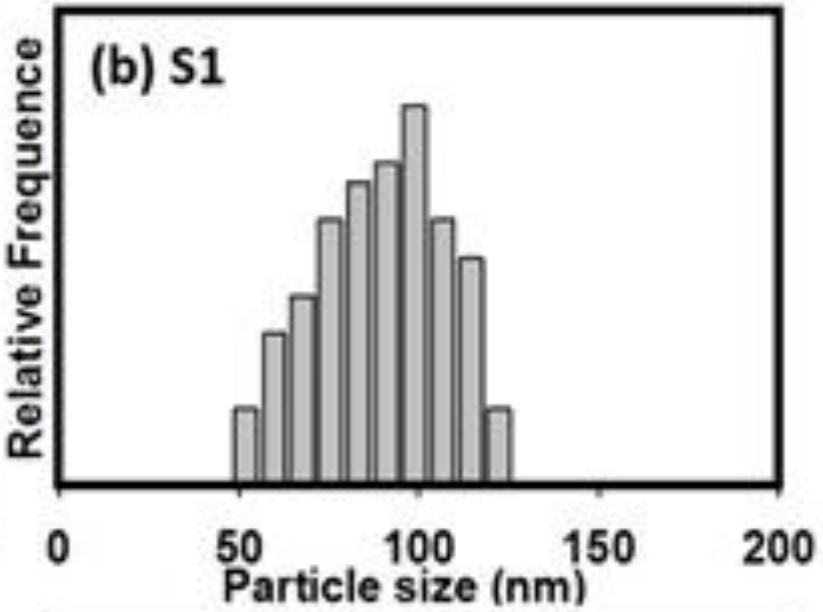
Scanning electron microscopy images of cube-like Fe3O4 nanoparticles with various particle sizes.

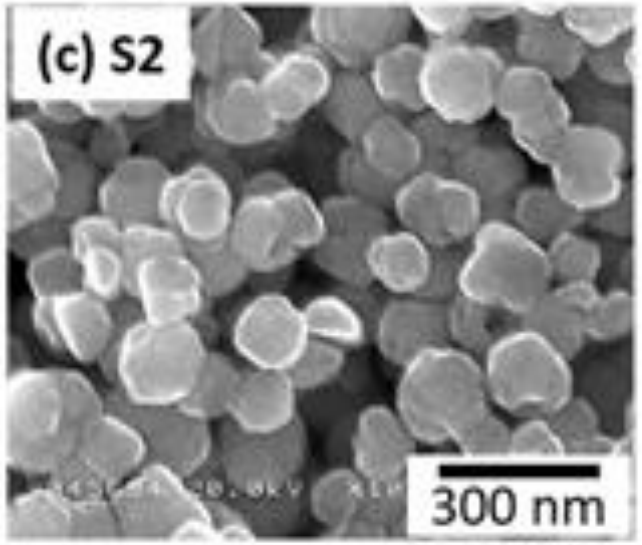
(a)9.6,
(b)19.6,
(c)24.4,
(d)31.9,
(e)45.3,
(f)64.7,
(g)130,
(h)243, and
(i)287 nm,

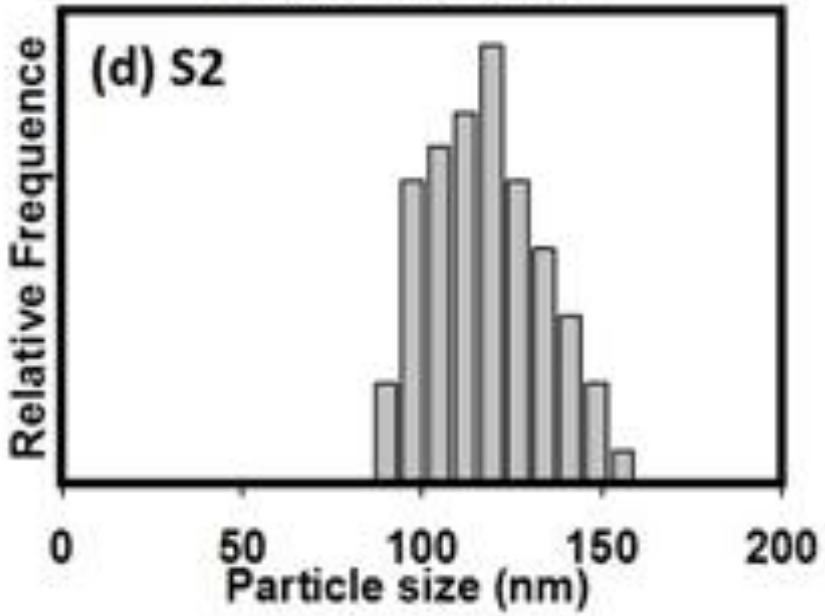
which are named as C1–C9 respectively.

Case study 3:









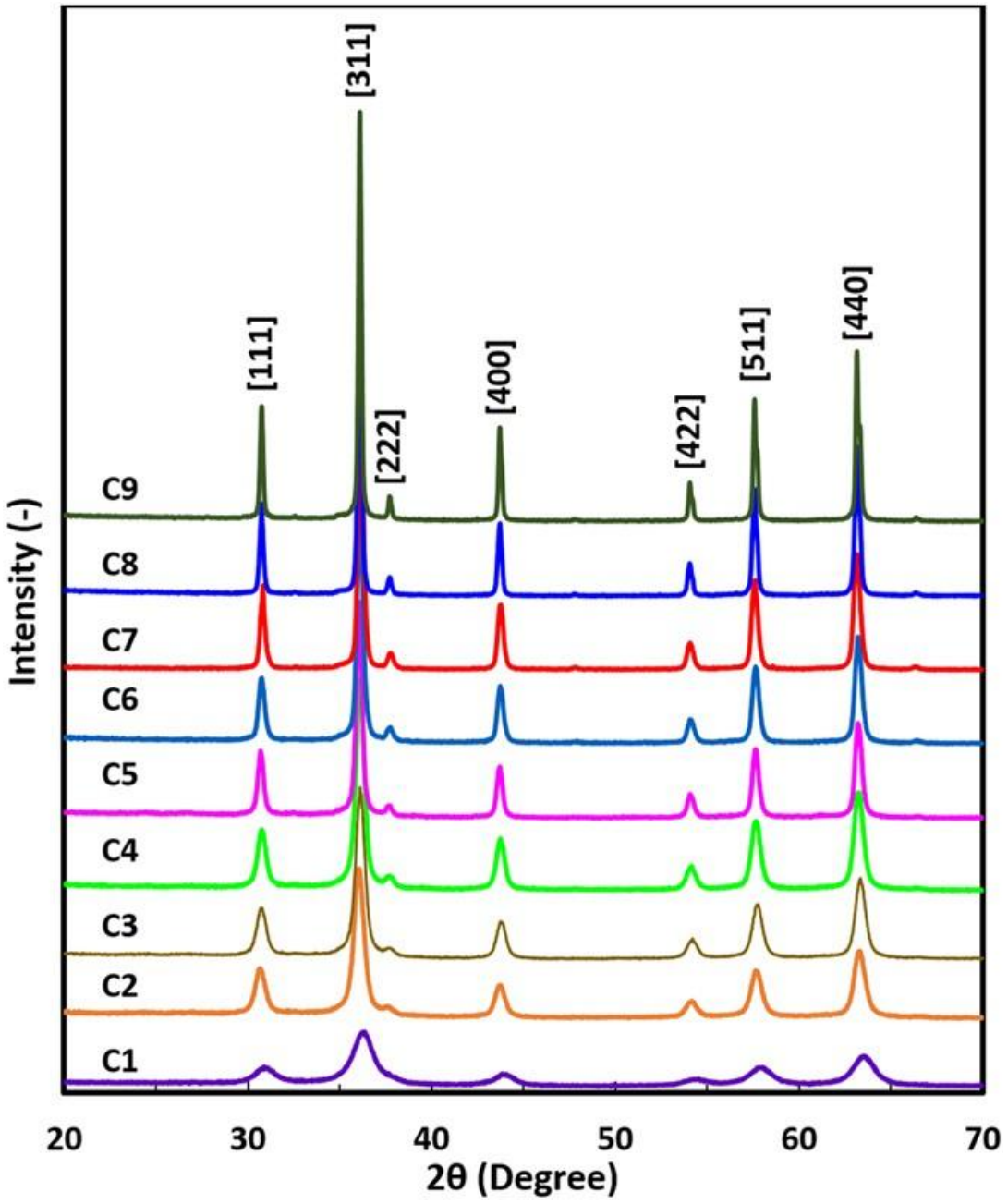
- Highly crystalline single-domain 10-300nm magnetite Fe3O4 nanoparticles have been synthesized using large-scale liquid precipitation method

Particle properties of sphere-like Fe_3O_4 nanoparticles:

S1 (93.3 nm) and
S2 (121 nm):

(a) and (c) scanning electron microscopy images,
(b) and (d) particle size distributions

Case study 3:



Compared to our previous case study:

- The sharp peaks indicate the highly crystalline nature of the Fe_3O_4 NPs.

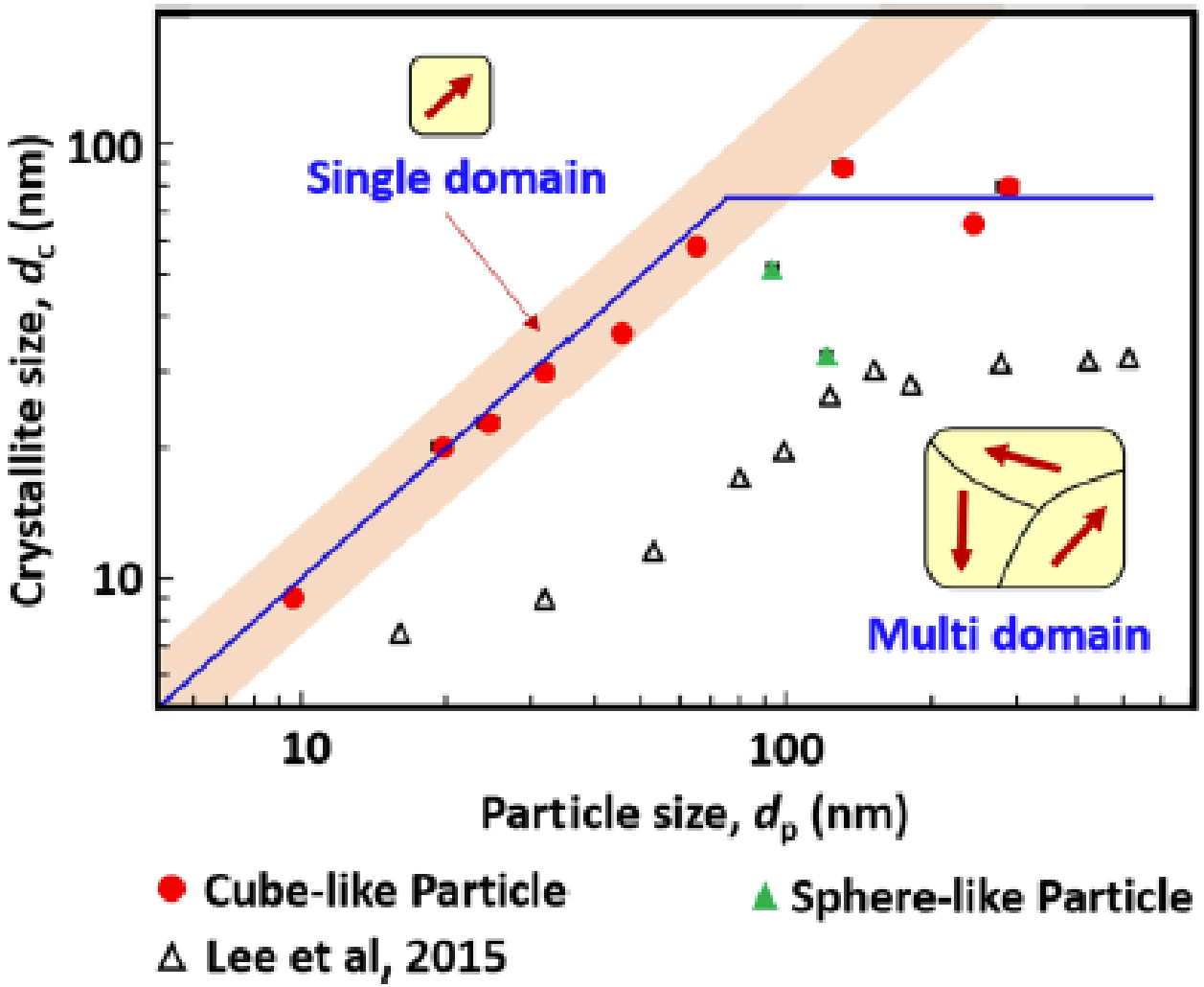
- Impurity peaks and transition phases were not observed, which indicates that the particles prepared by the liquid precipitation method were pure, both chemically and in their crystalline phase.

The crystallite sizes (d_c) of all the Fe_3O_4 NPs were estimated by the Scherrer formula using the highest intensity XRD peak, namely [311], and compared with the d_p obtained from the TEM analysis

Case study 3:

- Highly crystalline single-domain 10-300nm magnetite Fe3O4 nanoparticles have been synthesized using large-scale liquid precipitation method

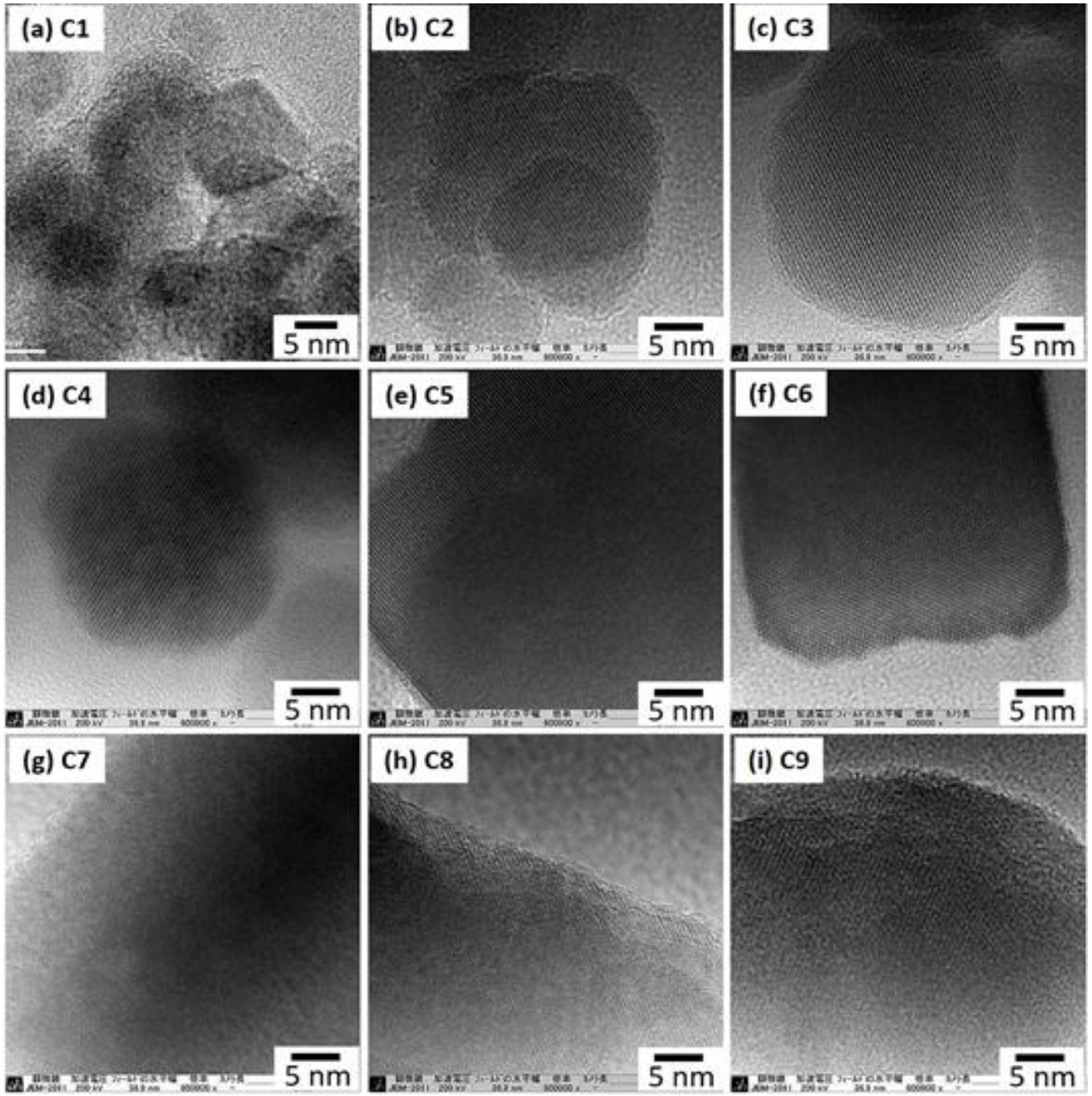
Sample	Particle size d_p (nm)	Crystal size d_c(nm)
C1	9.6	9.1
C2	19.6	20.2
C3	24.4	22.9
C4	31.9	29.9
C5	45.3	36.9
C6	64.7	58.2
C7	130	88.7
C8	243	65.8
C9	287	79.6
S1	93.3	51.7
S2	121	32.4



Particle size was calculated from SEM/TEM/Particle size analyser and crystallite size was calculated by Debye– Scherrer equation from XRD.

The straight line indicates a single crystalline particle, whereas the particles corresponding to points that fall below this line are polycrystalline. the dc and corresponding dp have almost the same value for particles with a diameter of 10 to around 80 nm, which indicates a single-domain structure. The particles with diameters of larger than 80 nm have a constant dc, which indicates a multi-domain structure. This can be explained the by the single and multi directional lattice fringes by TEM

Case study 3:

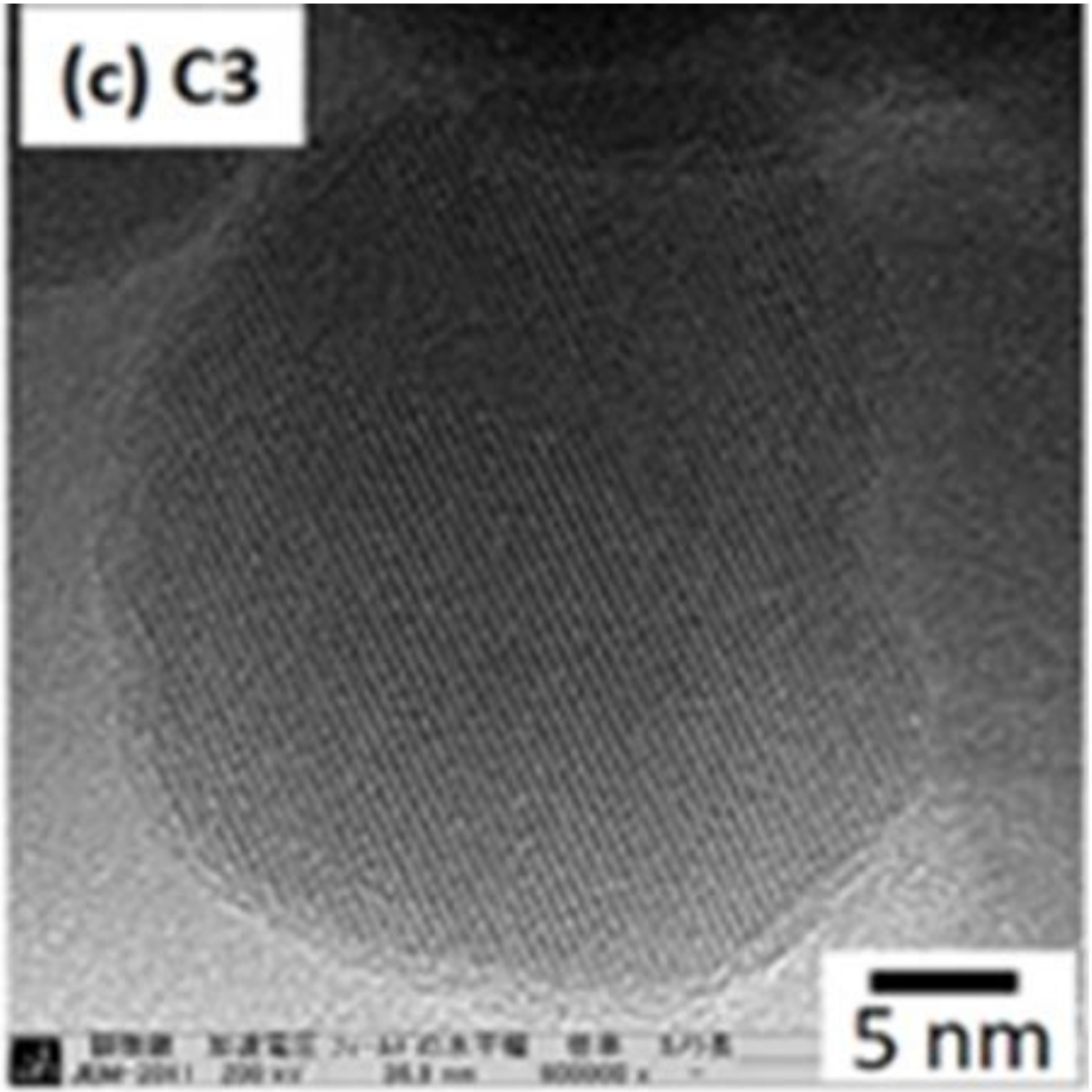


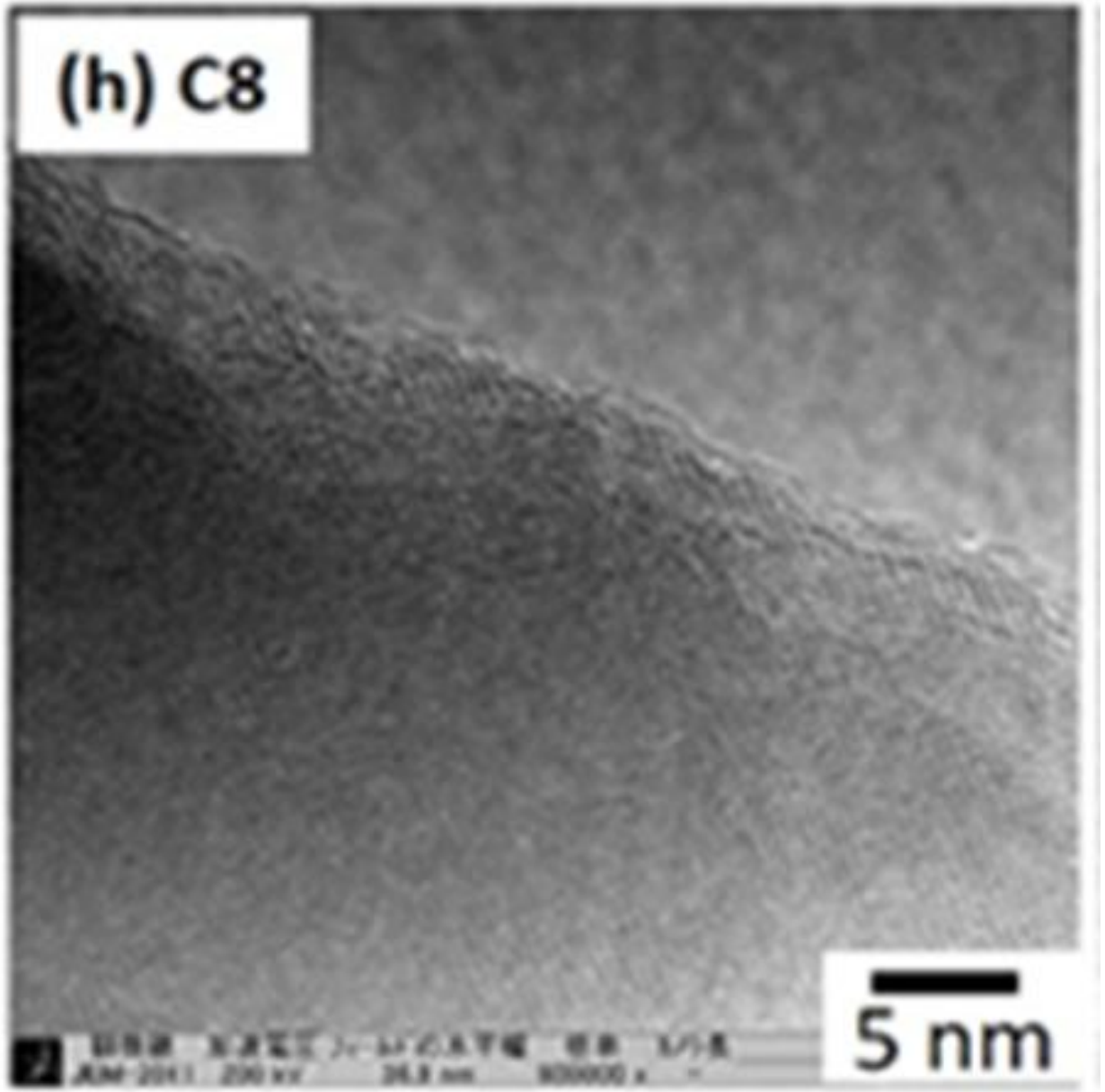
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High resolution transmission electron microscopy images of cube-like Fe_3O_4 nanoparticles with various sizes. (a) 9.6, (b) 19.6, (c) 24.4, (d) 31.9, (e) 45.3, (f) 64.7, (g) 130, (h) 243, and (i) 287 nm, which are named C1–C9, respectively.

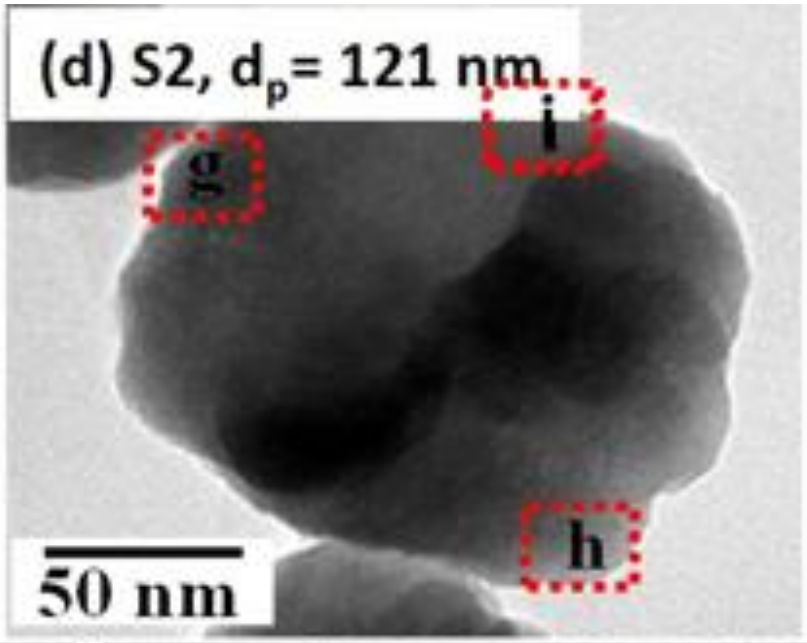
TEM shows the single direction lattice fringes of the NPs, and indicate that the NPs have a single crystalline structure up to a *dp* of 64.7 nm. In contrast, the C7–C9 NPs, with diameters larger than 100 nm, show a polycrystalline structure multi directional fringes.

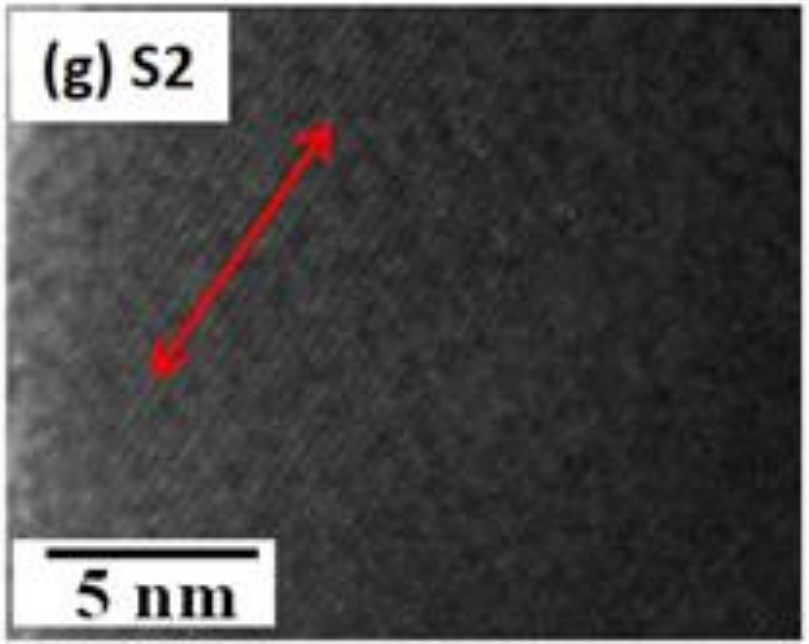
Case study 3:

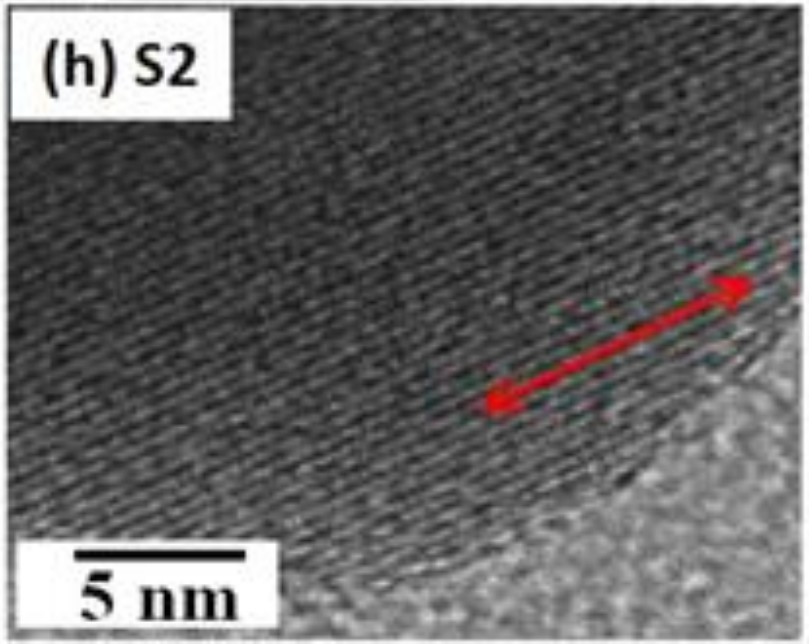


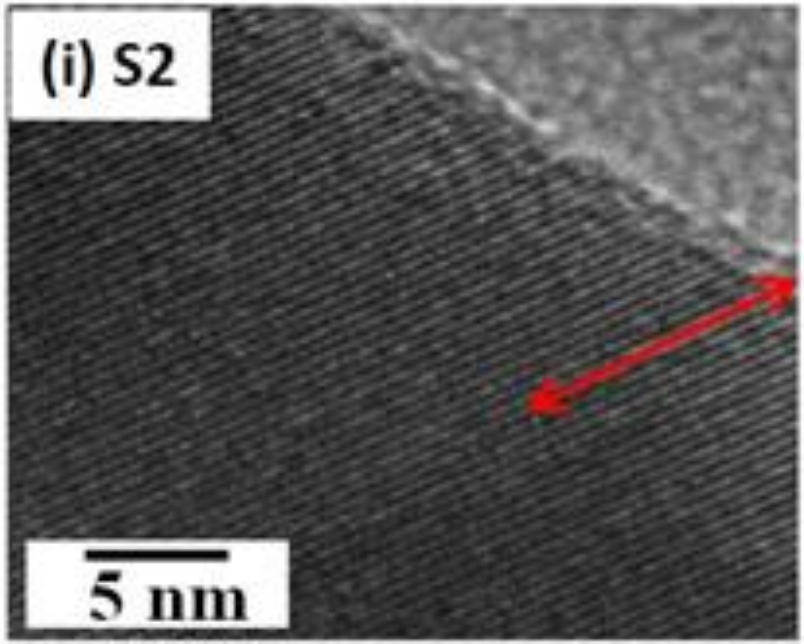


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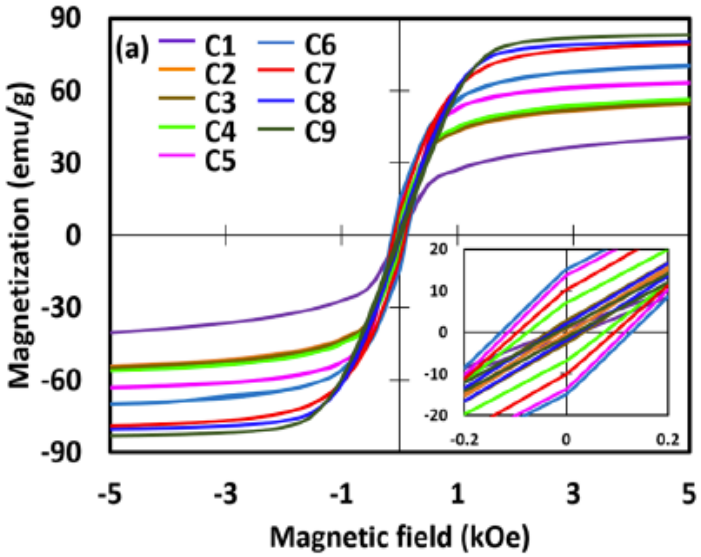


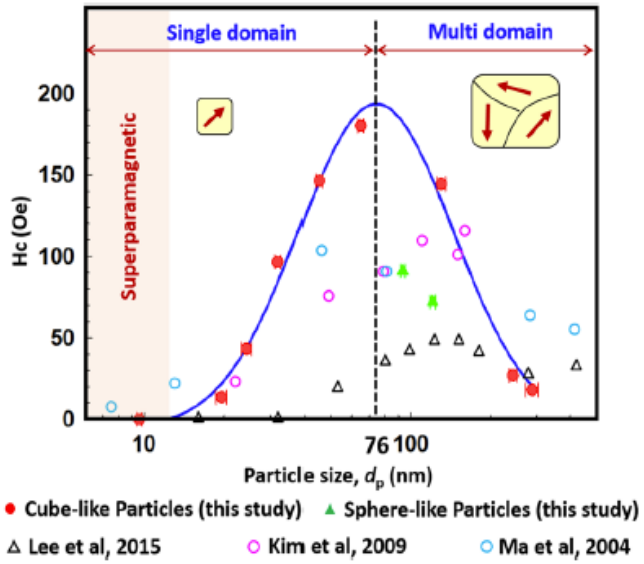


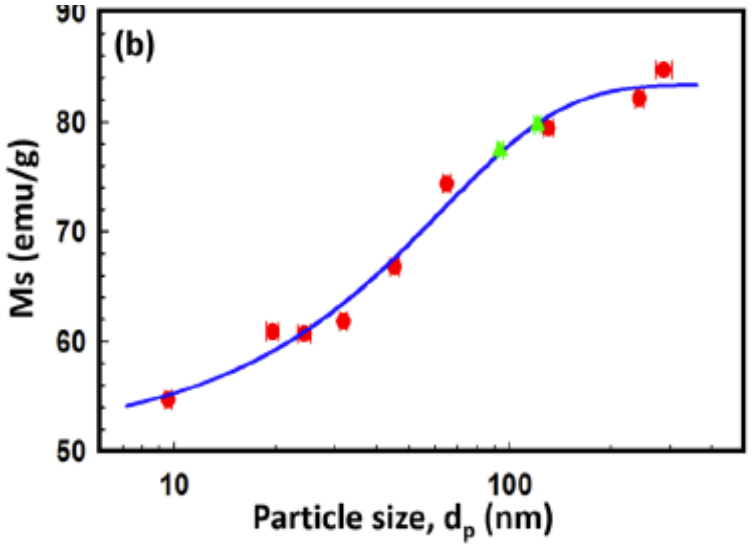


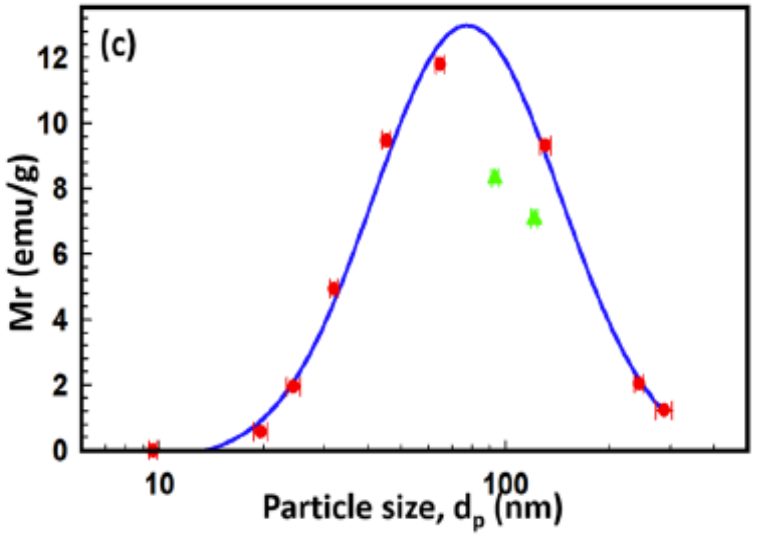
Case study 3:

- Highly crystalline single-domain 10-300nm magnetite Fe3O4 nanoparticles have been synthesized using large-scale liquid precipitation method









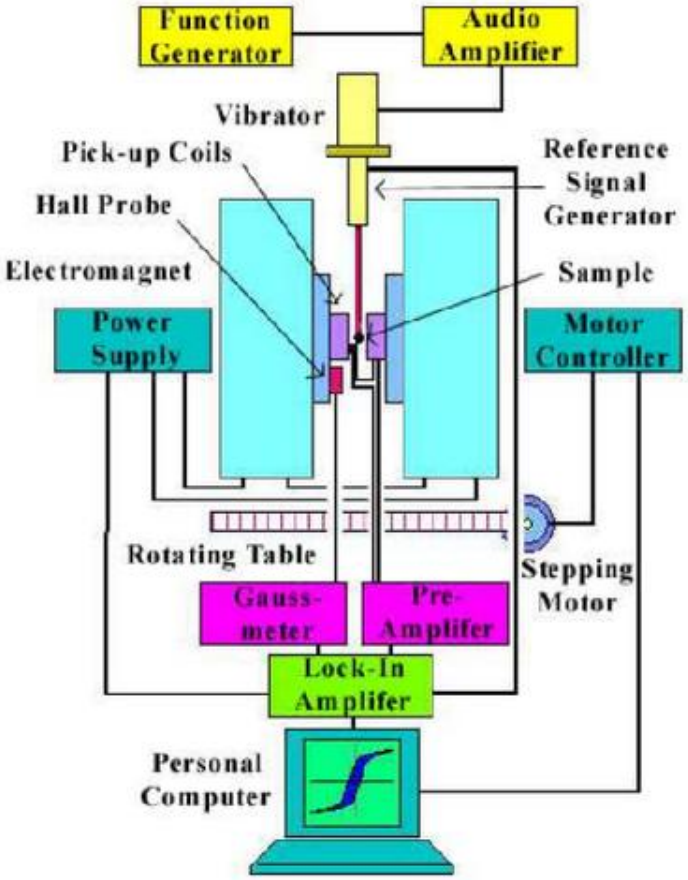
Cube-like Particles Sphere-like Particles

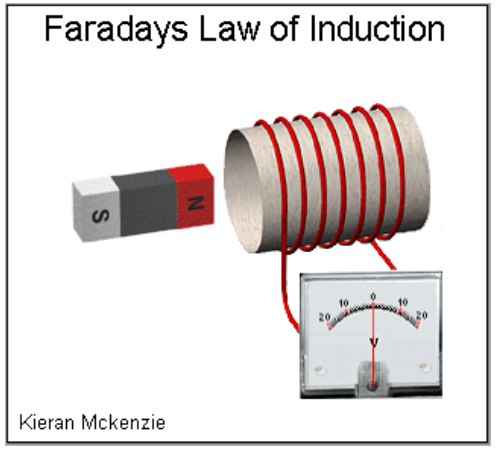
1. The hysteresis loops for these particles, show the ferrimagnetic nature.
2. The *Ms* increases with increasing particle size, regardless of the crystal structure and particle shape.
3. The *Hc* and remanent magnetisation (*Mr*), which are also affected by *d*p
4. Superparamagnetic when Mr = 0 and Hc =0, C1 samples is only superparamagnetic
5. 76nm (single domain) sample has highest Mr and Hc. After that Mr and Hc decreases even further increase of dp (multi domain)
6. Multiple orientation of the polycrystalline samples crystallites within the multi-domain-structured NPs is considered to be the main reason for their lower Mr and Hc.

These findings suggest that considerable attention should be given to the particle size and crystalline properties of Fe3O4 NPs, which have potential biomedical and industrial applications. These applications require that magnetic particles are sized appropriately to achieve a good balance between effective surface area and satisfactory magnetic performance.

Magnetic properties characterization techniques

Vibrating Sample Magnetometer (VSM)





https://www.daenotes.com/electronics/basic-electronics/faraday-laws-of-electromagnetic-induction



1. VSM works as Faraday's electromagnetic induction
2. A VSM is used to measure the magnetic behaviour of magnetic materials.
3. Placing the sample to be studied in a constant magnetic field (by electromagnet)
4. If the sample is magnetic, this constant magnetic field will magnetize the sample
5. The magnetic dipole moment of the sample will create a magnetic field around the sample, sometimes called the magnetic stray field

6. As the sample is moved up and down, this magnetic stray field is changing as a function of time and can be sensed by a set of pick-up coils.

7. The alternating magnetic field will cause an electric field in the pick-up coils according to Faraday's Law of Induction. This current will be proportional to the magnetization of the sample.

8. The induction current is amplified by a trans impedance amplifier and lock-in amplifier

9. system can tell you how much the sample is magnetized and how its magnetization depends on the strength of the constant magnetic field.

http://gato-docs.its.txstate.edu/jcr:7543b8ec-f619-49c3-888f-01c3ad13c7da/vsm.pdf

Magnetic properties characterization techniques

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Superconducting Quantum Interface Device (SQUID)



The superconducting quantum interference device (SQUID) are very sensitive magnetometers based on superconducting loops consists of two superconductors separated by thin insulating layers to form two parallel **Josephson junctions**.

SQUID magnetometer is one of the most effective and sensitive ways of measuring magnetic properties.

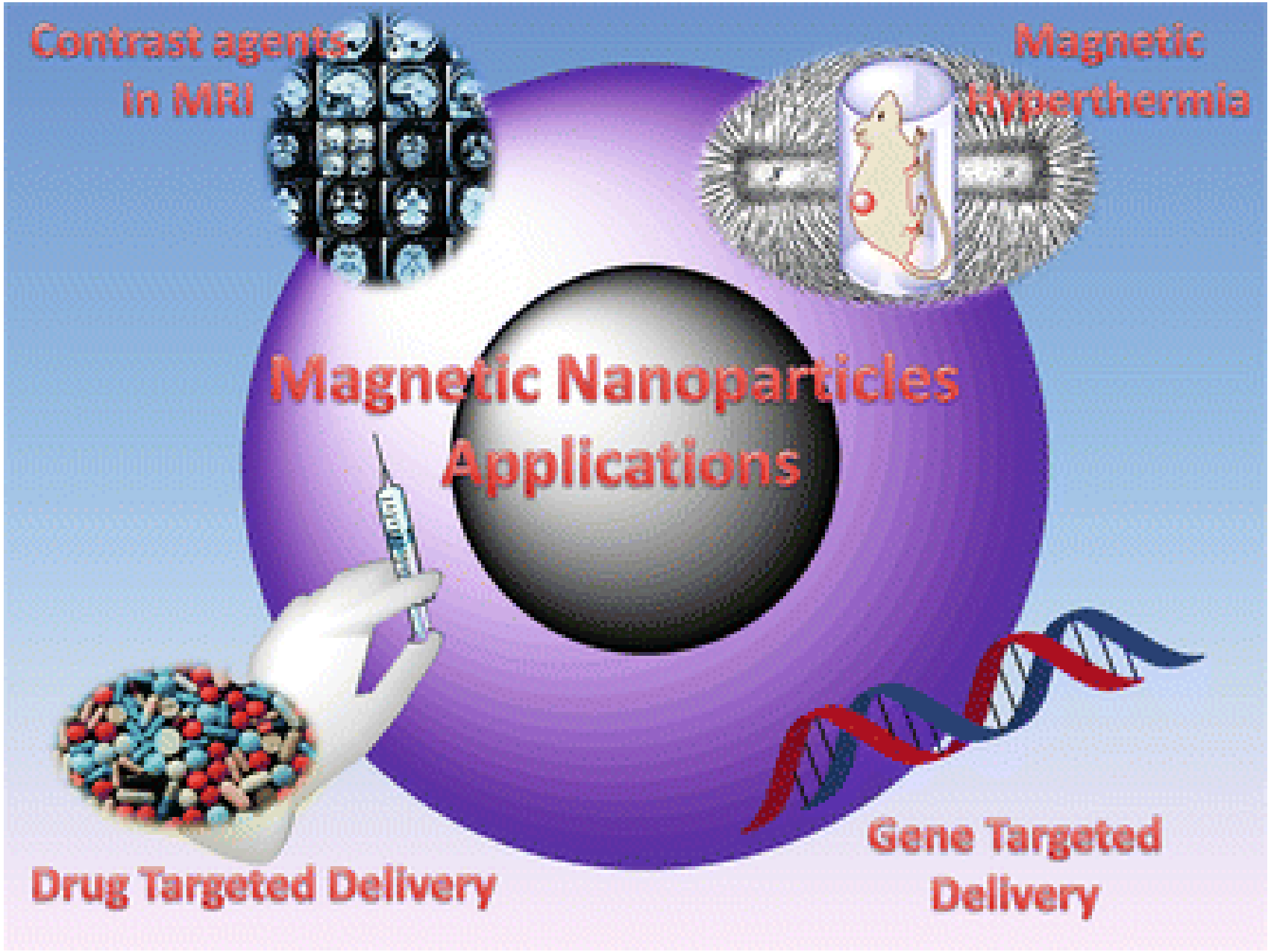
The device may be configured as a magnetometer to detect incredibly small magnetic fields small enough to measure the magnetic fields in living organisms.

Squids have been used to measure the magnetic fields in mouse brains to test whether there might be enough magnetism to attribute their navigational ability to an internal compass.

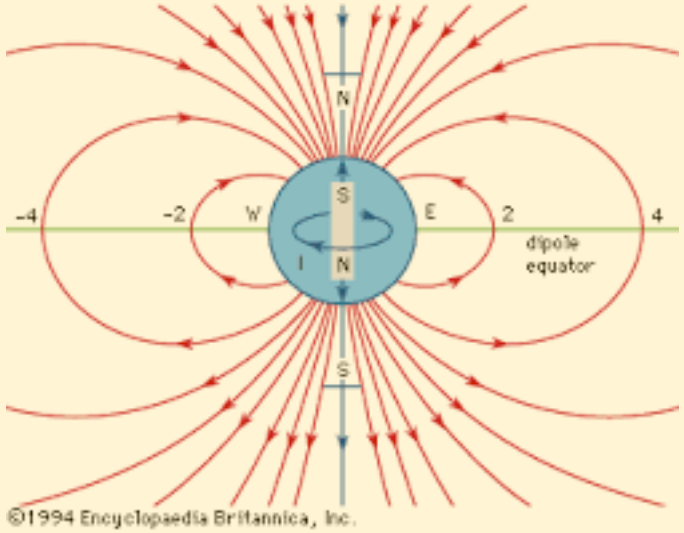
Cryocooler Compressor: The MPMS EverCool system features an integrated cryocooler–Dewar System that recondenses the He directly with in the EverCool Dewar.

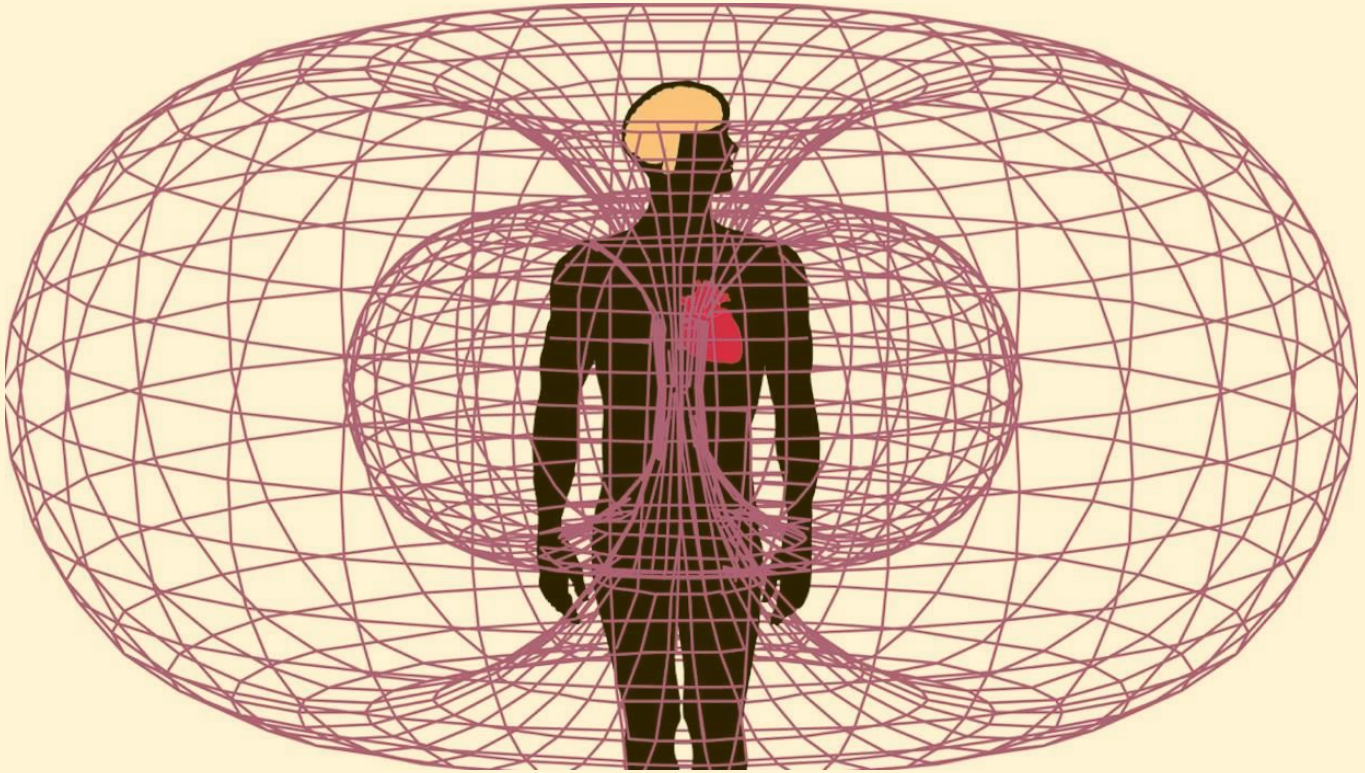
http://squid.iitd.ernet.in/Basic_Literature.htm

Applications – magnetic nanoparticles



Bringing the Magnetic field into Biomedical Applications





Applications – Nano-machineries

DARLINGHURST

Forces fight for a cure

A NEW method of targeting tumours using magnetic forces could improve the treatment and wellbeing of cancer patients.

Researchers at Darlinghurst's Victor Chang Cardiac Research Institute and the University of Wollongong are developing the method that will enable doctors to deliver cancer medications directly to a tumour.

The proposed treatment involves attaching tiny magnetic particles to drug carriers that can be guided through the body to the cancer using a magnetic field. A magnetic field will also be used to release the drugs directly to the cancer cells.

Medications ingested for conventional cancer treatments disperse through the bloodstream and attach to the cancer cells where they interfere with the molecules responsible for its growth.



Dr Shahriar Hossain and Prof Boris Martinac are developing tumours targeting using a magnetic method.

MAGNETIC CANCER TREATMENT

Large doses of highly toxic cancer medications are required to kill a tumour, often leading to serious side effects and damaging healthy tissues and organs.

Prof Boris Martinac of the Victor Chang Institute, said magnetically targeted drug delivery would help to reduce the amount of cancer-killing medication needed to kill the tumour and would reduce side effects.

"We want to attach tiny magnetic particles to cancer drugs. The doctor can then direct the drugs through the patient's bloodstream to the tumour using a handheld magnetic device that emits a magnetic field, but is not harmful for the patient," he said.

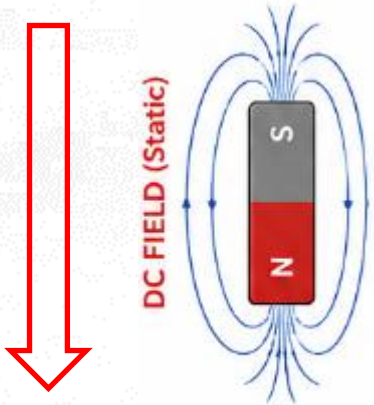
"When the drugs reach the tumour different alternating magnetic forces can be applied to the particles, causing the protein molecules to open, much like a valve or turning on a tap, releasing the drug that will then attack the cancer cells.

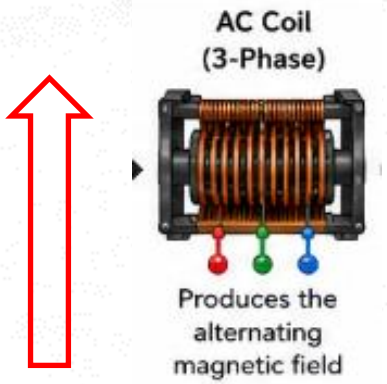
He said it was exciting because it could potentially reduce the need for current treatments such as chemotherapy and surgery.

UOW materials scientist and Australian Research Council Discovery Early Career Researcher Award Fellow, Dr Shahriar Hossain and his team, will build a device that doctors could use to "drive" the drugs through the body.

Magneto-Mechanical Nano-machineries for Precision Drug Release

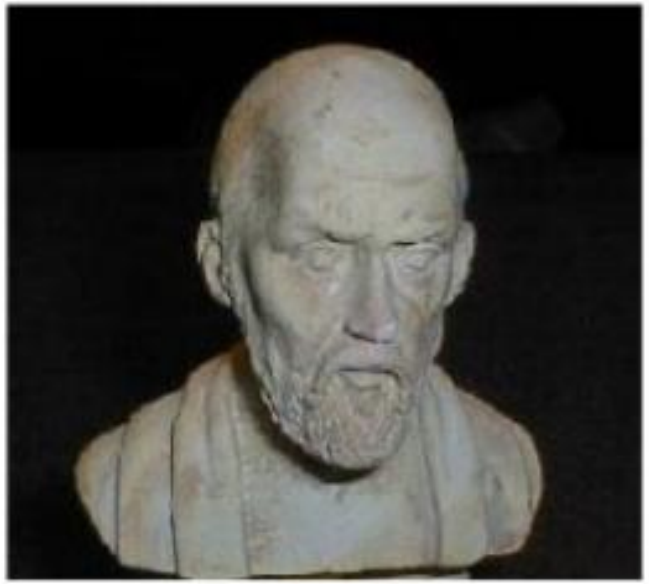
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AUSTRALIA





Applications – Magnetic Hyperthermia – Nano-heaters

The University Of Queensland AUSTRALIA



"Those diseases which medicines do not cure,
the knife cures;
those which the knife cannot cure, fire cures;
and those which fire cannot cure,
are to be reckoned wholly incurable."

—Hippocrates (460-370 BC)

- Physical properties which make magnetic materials attractive for biomedical applications are,
  - First, that they can be manipulated by an external magnetic field – this feature is useful for separating, immunoassay and drug targeting, and
  - Second, hysteresis and other losses occur in alternating magnetic fields – this is useful in hyperthermia applications.

Applications – Hyperthermia – Nano-heaters

- **Hyperthermia** is an elevated body temperature due to failed thermoregulation. Hyperthermia occurs when the body produces or absorbs more heat than it can dissipate.
- Hyperthermia can be created artificially by drugs or medical devices. Hyperthermia therapy may be used to treat some kinds of cancer and other conditions, most commonly in conjunction with radiotherapy. It is known that cancer cells show more weakness to high temperatures compared with ordinary cells.
- Hyperthermia system that is able to heat tumor cells above 43°C without damaging normal tissues is required.

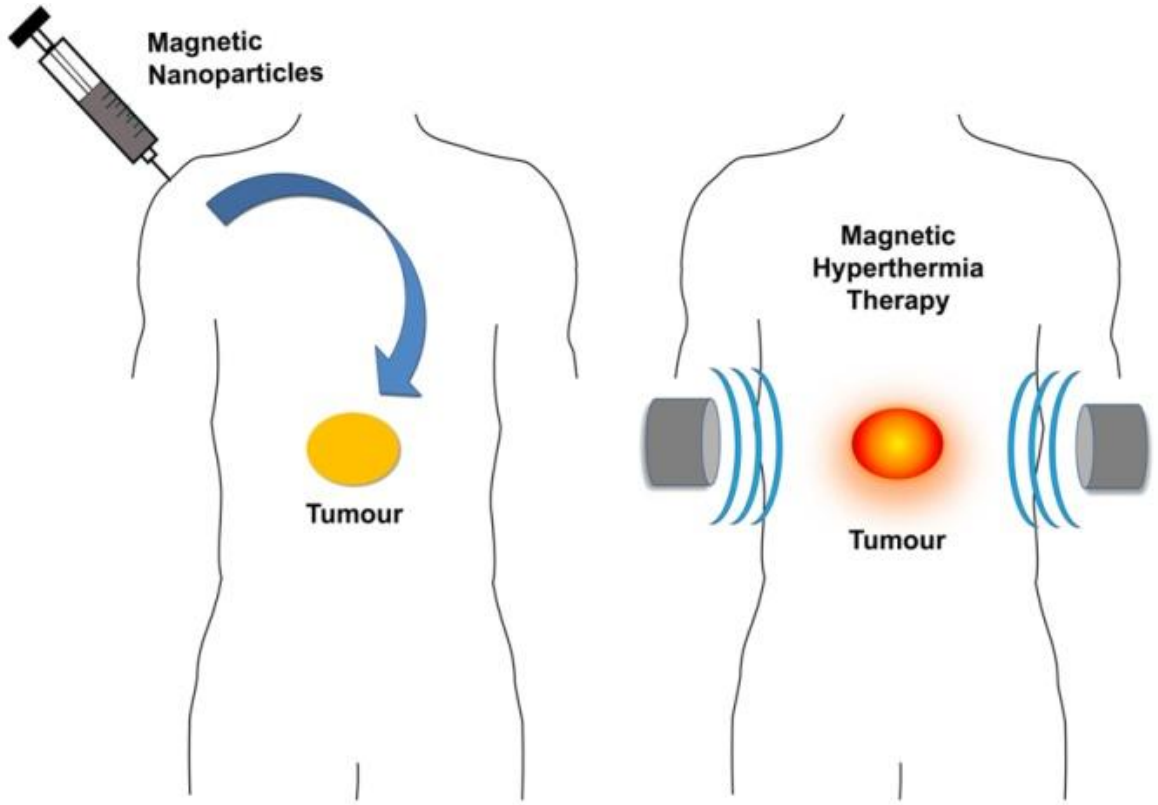
Applications – Hyperthermia – Nano-heaters



Applications – Hyperthermia – Nano-heaters

The magnetic moments on nanoparticles will align with an external magnetic field. As the external field changes direction, the magnetic moment will also change direction. This produces dissipation leading to heating. One of the major advantages of using magnetic fields to produce heating is that they readily penetrate biological tissues

However, heating the surrounding tissue can also produce unwanted side-effects so there are advantages strictly controlling the region under treatment by using magnetic nanoparticles as the heating element. Superparamagnetic nanoparticles absorb power from AC magnetic fields and heat surrounding tissues.



https://www.intechopen.com/books/biomedical-engineering-frontiers-and-challenges/coating-nanomagnetic-particles-for-biomedical-applications

Applications – Hyperthermia – Nano-heaters

The University Of Queensland AUSTRALIA

This thermotherapy is based on the fact that magnetic nanoparticles can transform electromagnetic energy from an external high-frequency field to heat.

As a consequence, if magnetic nanoparticles are put inside a tumor and the whole patient is placed in an alternating magnetic field of well-chosen amplitude and frequency, only the tumor temperature would rise. The elevation of temperature may enhance tumor oxygenation , radio- and chemosensitivity, thus shrinking tumors.

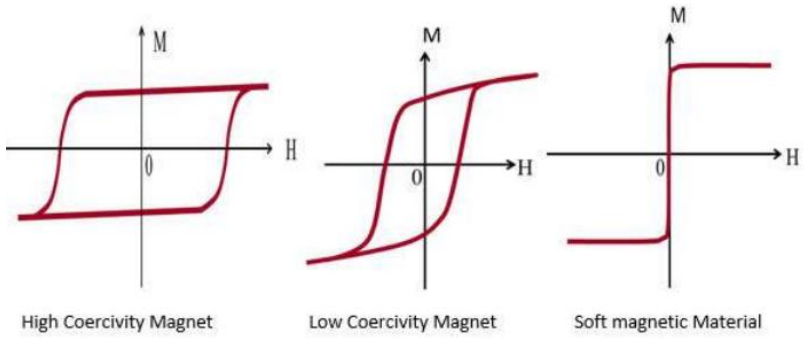
Difficulty in heating the local tumor region WITHOUT DAMAGING normal tissue

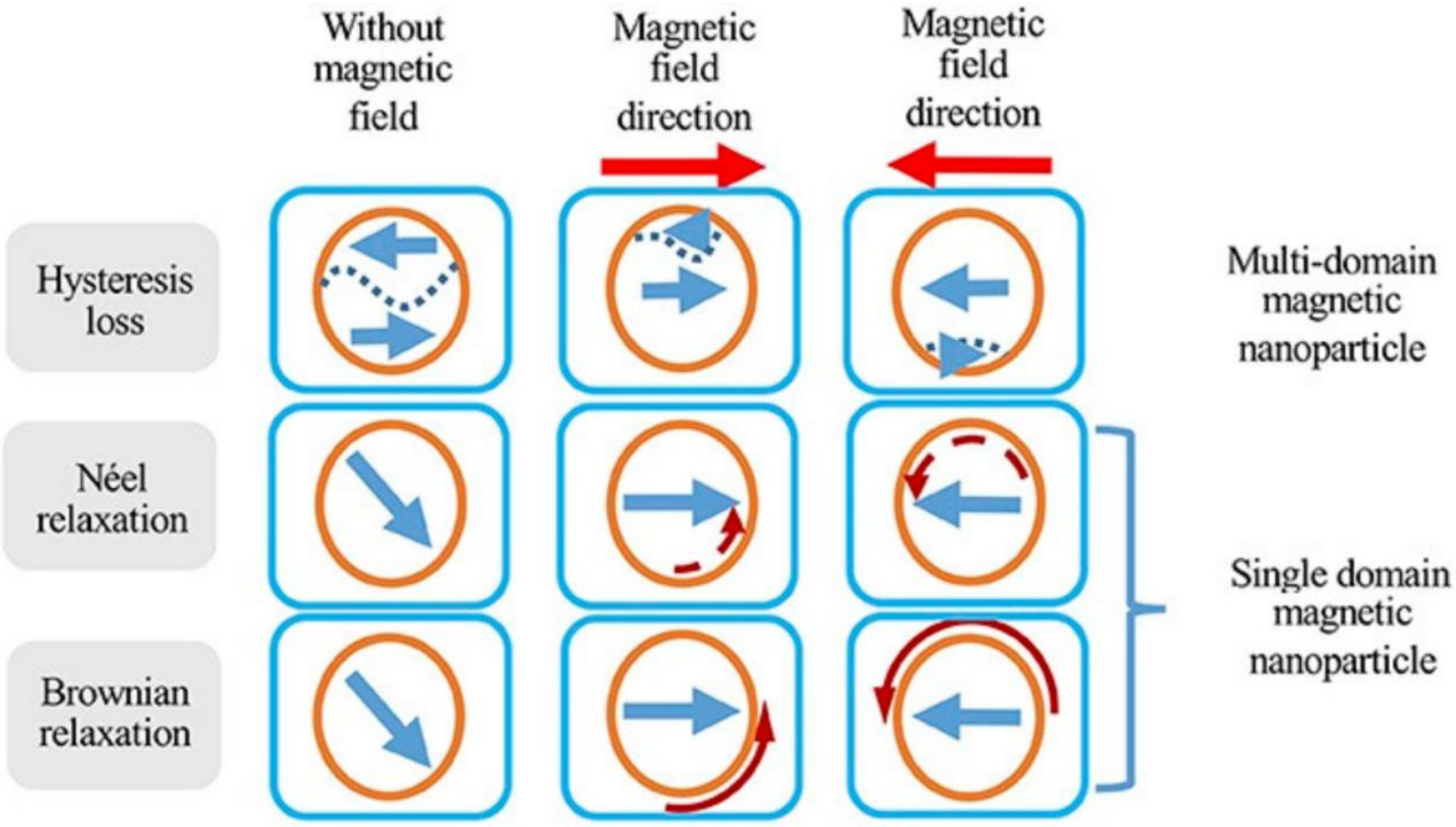
Applications – Hyperthermia – Nano-heaters

What are the mechanism?

Heat dissipation from the magnetic nanoparticles under an AC field is caused by

1)Brownian relaxation loss
2)Neel's spin relaxation
3)Hysteresis loss and
4)Eddy current loss.

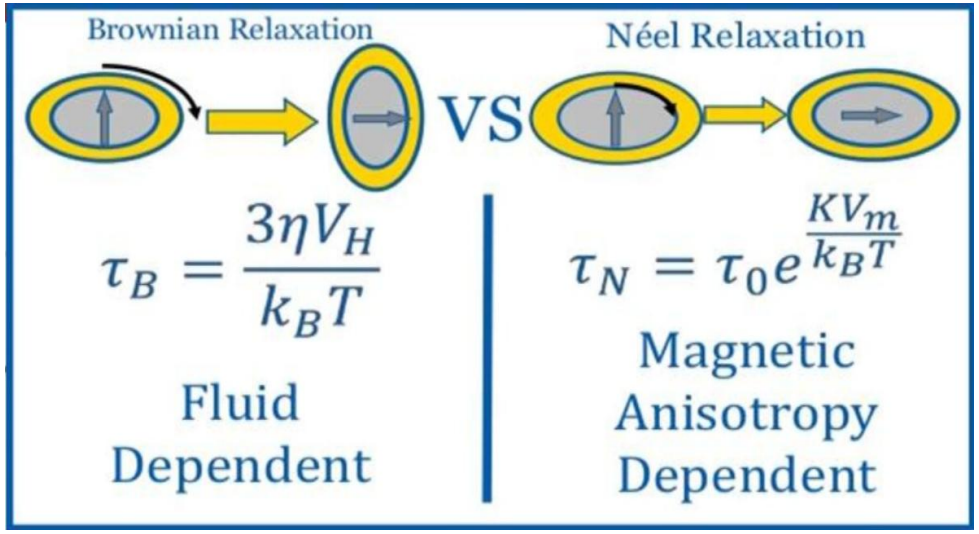




https://www.mdpi.com/1424-8220/20/9/2554/htm

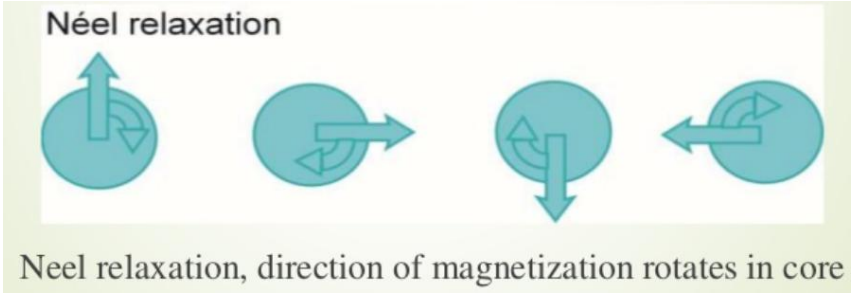
Magnetic Particle Relaxation

The well dispersed magnetic nanoparticles in a fluid would show two types of relaxations, ***Brownian relaxation*** (particle rotation, occurs in all types of nanoparticles) and ***Neel's spin relaxation*** (magnetic moment rotation, occurs only in superparamagnetic nanoparticles where no physical rotation of the particle is possible).



η is the dynamic viscosity of carrier liquid and
V is the hydrodynamic radius of the particle,
k_B is Boltzmann constant,
$\tau 0 = 10^{-9}$ s





Case study problem solving

Two magnetite nanoparticles are being considered for biomedical sensing:

Particle S (small core, coated): core diameter $D_c = 8\ \mathrm{nm}$; hydrodynamic diameter $D_h = 12\ \mathrm{nm}$.
Particle L (large core, lightly coated): core diameter $D_c = 20\ \mathrm{nm}$; hydrodynamic diameter $D_h = 22\ \mathrm{nm}$.

Use:

- $K = 1.0 \times 10^4\ \mathrm{J\,m^{-3}}$
- $k_B = 1.38 \times 10^{-23}\ \mathrm{J\,K^{-1}}$
- $T = 300\ \mathrm{K}$
- $\tau_0 = 1 \times 10^{-9}\ \mathrm{s}$
- viscosity $\eta = 1.0 \times 10^{-3}\ \mathrm{Pa \cdot s}$
- sphere volume $V = \frac{4}{3}\pi\left(D_c/2\right)^3$,hydrodynamic volume $V_h = \frac{4}{3}\pi\left(D_h/2\right)^3$
- Néel relaxation with $E_B = KV$
- Brownian relaxation $\tau_B = 3\eta V_h/(k_B T)$
- Effective relaxation $\tau_{eff} = \frac{\tau_N \tau_B}{\tau_N + \tau_B}$.

Decide for each particle: (a) is it superparamagnetic (use $E_B$vs $25k_BT$ as the blocking criterion)? (b) which relaxation dominates (shorter time)?

Case study problem solving - answer

Common numbers

- $k_B T = 1.38 \times 10^{-23} \times 300 = 4.14 \times 10^{-21}$ J
- $25 k_B T = 1.035 \times 10^{-19}$ J(used as blocking threshold)

Particle S (8 nm core, 12 nm hydrodynamic)

1.Core volume

$r = 4\ \text{nm} = 4.0 \times 10^{-9}\ \text{m}\rightarrow$

$$V = \frac{4}{3}\pi r^3 = 4.18879 \times \left(4.0 \times 10^{-9}\right)^3 = 4.18879 \times 64 \times 10^{-27} = 2.6808 \times 10^{-25}\ \text{m}^3.$$

2.Energy barrier

$$E_B = KV = 1.0 \times 10^4 \times 2.6808 \times 10^{-25} = 2.68 \times 10^{-21}\ \text{J}.$$

3.Superparamagnetic check

$E_B = 2.68 \times 10^{-21} \ll 25 k_B T = 1.035 \times 10^{-19} \rightarrow$**superparamagnetic** (thermally unstable / flips easily).

4.Néel relaxation time

exponent $= E_B/(k_B T) = 2.68 \times 10^{-21}/4.14 \times 10^{-21} \approx 0.647.$

$$\tau_N = 10^{-9}\, e^{0.647} \approx 1.91 \times 10^{-9}\ \text{s}.$$

5.Hydrodynamic volume (Dh = 12 nm → rh = 6 nm)

$$V_h = 4.18879 \times \left(6.0 \times 10^{-9}\right)^3 = 6.556 \times 10^{-23}\ \text{m}^3.$$

6.Brownian relaxation time

$$\tau_B = \frac{3\eta V_h}{k_B T} = \frac{3 \times 10^{-3} \times 6.556 \times 10^{-23}}{4.14 \times 10^{-21}} \approx 6.0 \times 10^{-7}\ \text{s}.$$

7.Effective relaxation & dominant mechanism

$\tau_N \approx 1.9 \times 10^{-9}$ svs $\tau_B \approx 6.0 \times 10^{-7}$ s.

→ **Néel much faster** → **Néel-dominated** (internal flips dominate). $\tau_{eff} \approx \tau_N \approx 2 \times 10^{-9}$s.

Case study problem solving

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Particle L (20 nm core, 22 nm hydrodynamic)

1.Core volume

$r = 10\ \text{nm} = 1.0 \times 10^{-8}\ \text{m}\rightarrow$

$$V = 4.18879 \times \left(1.0 \times 10^{-8}\right)^3 = 4.18879 \times 10^{-24}\ \text{m}^3.$$

2.Energy barrier

$$E_B = KV = 1.0 \times 10^4 \times 4.18879 \times 10^{-24} = 4.19 \times 10^{-20}\ \text{J}.$$

3.Superparamagnetic check

$E_B = 4.19 \times 10^{-20} < 25k_BT = 1.035 \times 10^{-19}$ →still **technically superparamagnetic** by the $25k_BT$ rule (i.e., thermally able to flip), but the energy barrier is substantially larger than for Particle S.

4.Néel relaxation time

exponent $= E_B/(k_BT) = 4.19 \times 10^{-20}/4.14 \times 10^{-21} \approx 10.12.$

$\tau_N = 10^{-9}\, e^{10.12} \approx 2.48 \times 10^{-5}$ s(≈ 25 μs).

1.Hydrodynamic volume (Dh = 22 nm → rh = 11 nm)

$$V_h = 4.18879 \times \left(11.0 \times 10^{-9}\right)^3 \approx 4.04 \times 10^{-24}\ \text{m}^3.$$

2.Brownian relaxation time

$\tau_B = \frac{3\eta V_h}{k_BT} = \frac{3\times10^{-3}\times4.04\times10^{-24}}{4.14\times10^{-21}} \approx 2.93 \times 10^{-6}$ s (≈ 3 μs).

(Depending on rounding, values ≈ 3–4 μs are typical.)

3.Effective relaxation & dominant mechanism

$\tau_N \approx 25 \times 10^{-6}$ svs $\tau_B \approx 3 \times 10^{-6}$ s.

→ **Brownian is faster** here → **Brownian-dominated**. τ_{eff}≈a few μs (roughly the smaller of the two, slightly reduced by combination).

Case study problem solving

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Interpretation & practical meaning (summary)

τ_0 vs τ_n:

- τ_0 (here 10^{-9}s) is the intrinsic attempt time — same for both particles in the example.
- τ_n (Néel relaxation) depends strongly on particle **core volume**. Particle S: τ_n ~ 2 ns. Particle L: τ_n ~ 25 µs. So even with same τ_0, τ_n can differ by many orders of magnitude.

Which relaxation dominates?

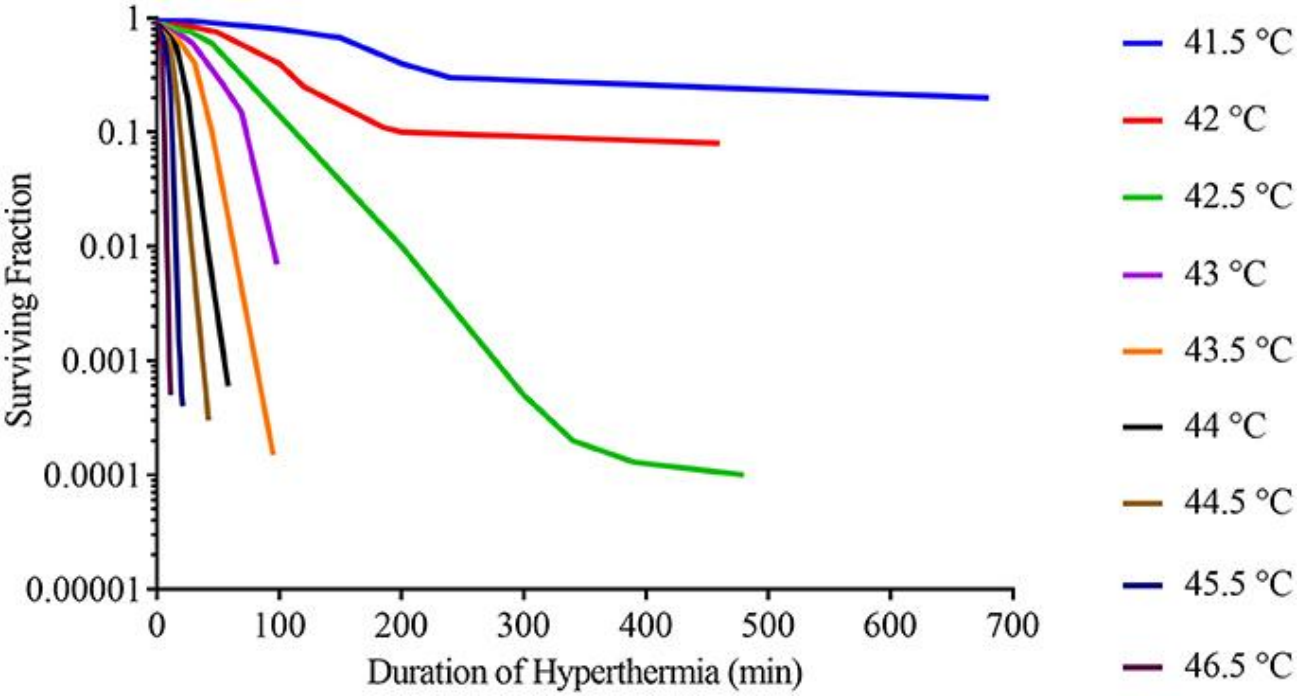
- **Particle S (8 nm core, Dh12):** Néel dominates. Magnetic response comes from internal moment flips — **good** if particles are immobilised (e.g., bound to cells), or if very fast magnetic response is needed.

- **Particle L (20 nm core, Dh22):** Brownian dominates. Magnetic response depends on physical rotation in fluid — **sensitive to viscosity and binding** (if it binds/immobilises, Brownian will be suppressed and the magnetic response will drop).

Superparamagnetism (blocking):

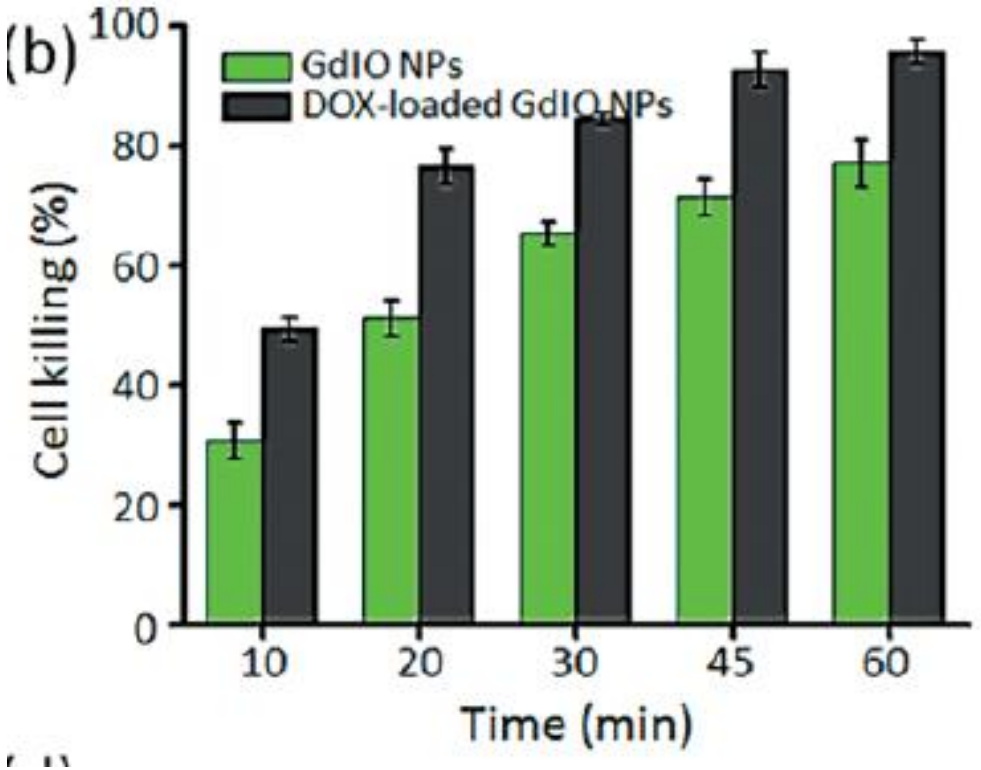
- Both particles have $E_B < 25k_BT$in this example → both are in the superparamagnetic regime by that rule. But their τ_n values differ: Particle S flips on nanosecond timescale, Particle L on microsecond timescale. Whether a particle appears "blocked" depends also on the measurement timescale (e.g., an experimental probe operating at 1 kHz sees dynamics on ~10^{-3} s — both here would appear superparamagnetic; a probe at MHz frequencies might respond differently).

Applications – Hyperthermia – Nano-heaters



https://www.frontiersin.org/articles/10.3389/fphar.2018.00831/full

Survival curves for asynchronous Chinese hamster ovary (CHO) cells heated at different temperatures for varying lengths of time



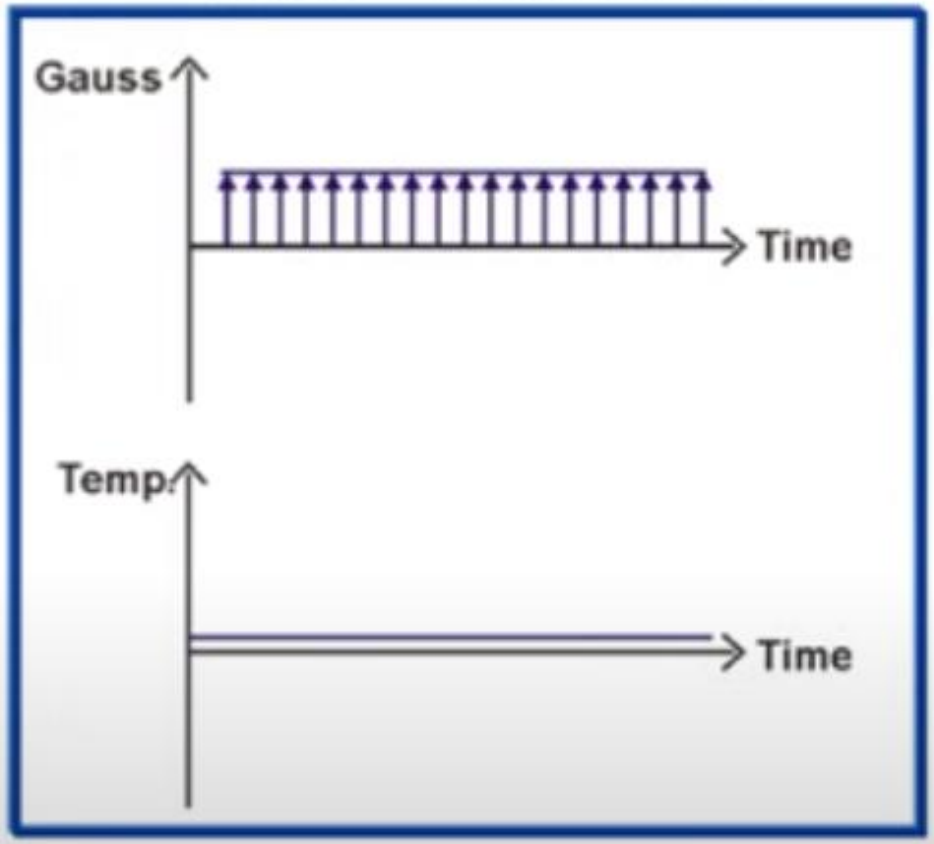
DOI: 10.1002/ejic.201600706

Cancercell killing by magnetic hyperthermia generated by bare GdIO NPs and DOX-GdIO NPs. The cells were treated with bare GdIO NPs or DOX-loaded GdIO NPs for 10–60 min, followed by the application of an alternating magnetic field (300 Oe, 267 kHz). After treatment, the cells were incubated under culture conditions for 24 h.

Applications – Magnetic Drug Delivery

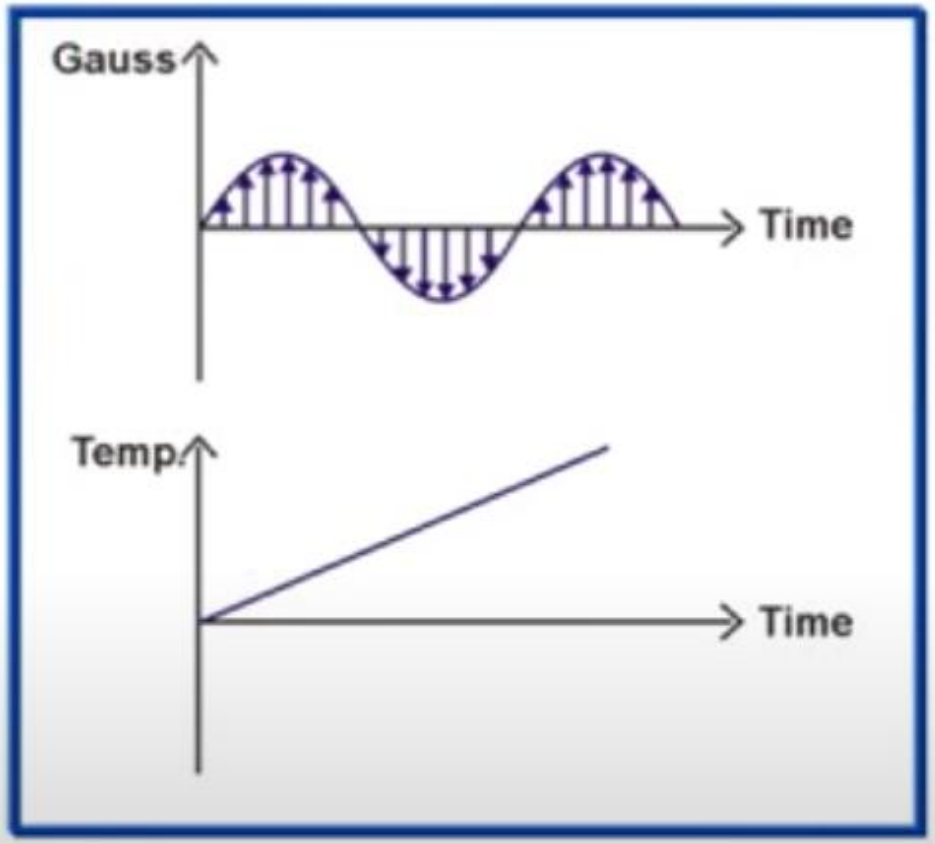
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Static – time-invariant



Very useful for imaging
Magnetic resonance imaging
(MRI)

Time-varying - alternating



Useful for therapy
Heat or activating devices

Applications – Magnetic Drug Delivery

Magnetically (DC/AC) triggered site specific drug delivery system using Fe_3O_4 as drug carrier

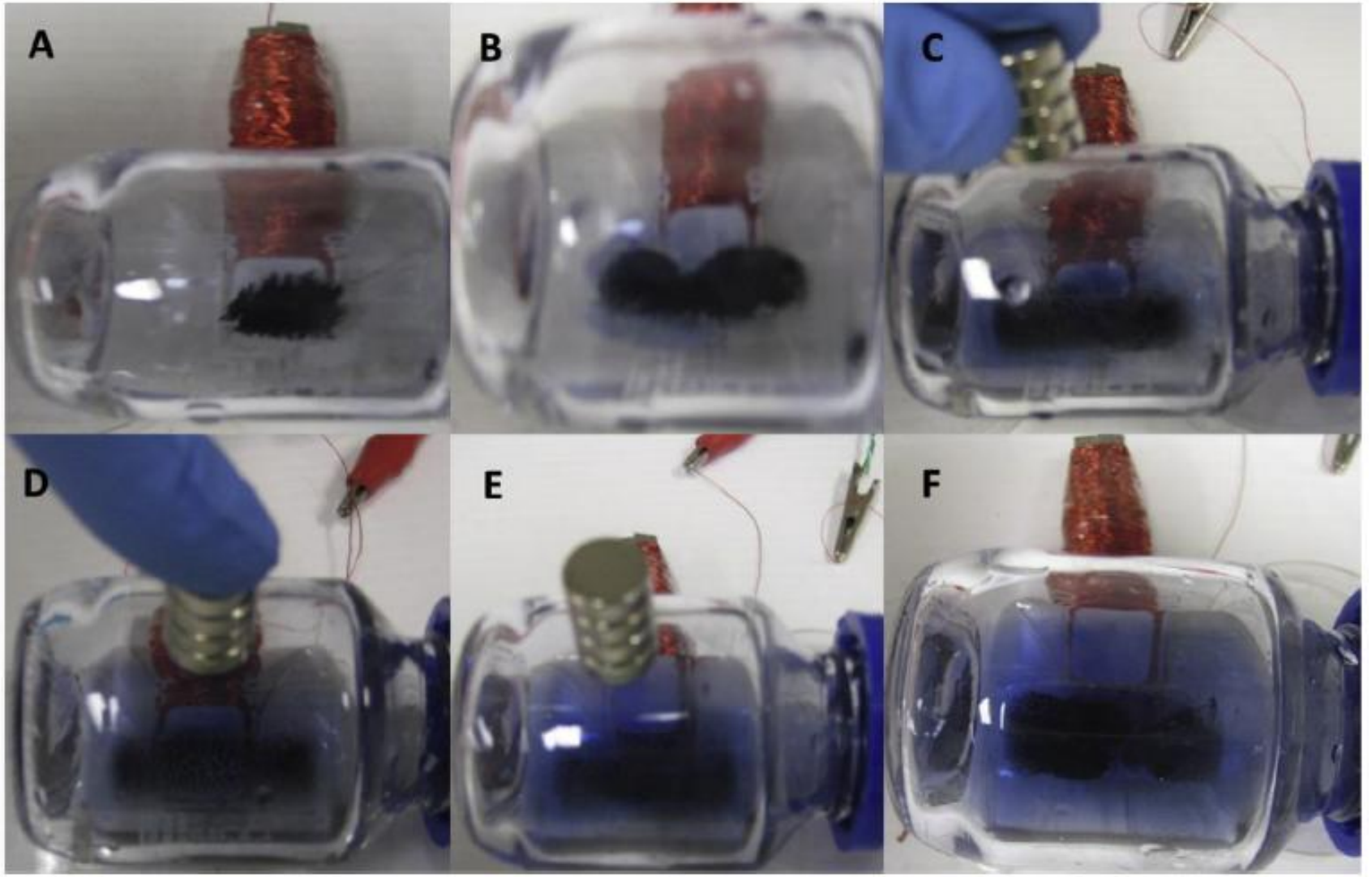


AC field only
(Is it localized or site specific?)

Applications – Magnetic Drug Delivery

Magnetically (DC/AC) triggered site specific drug delivery system using Fe_3O_4 as drug carrier

Site specific drug delivery system through combined AC and DC magnetic field



Ref. http://dx.doi.org/10.1016/j.micromeso.2015.12.032

Applications – Magnetic Drug Delivery

Magnetically (DC/AC) triggered site specific drug delivery system using Fe_3O_4 as



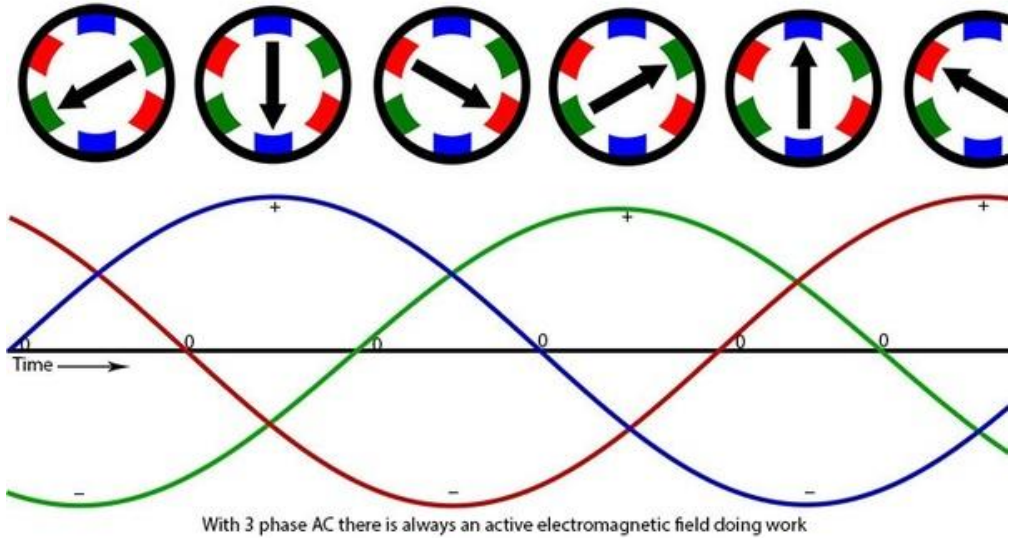
Ref. http://dx.doi.org/10.1016/j.micromeso.2015.12.032

Applications – Magnetic Drug Delivery

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Magnetically (DC/AC) triggered site specific drug delivery system using Fe_3O_4 as drug carrier

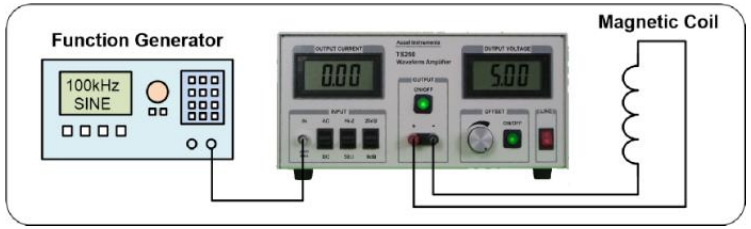
DC and AC magnetic field: Why do you need these two types?





DC field

Site specific targeted zone



AC field

Drug Release

AC field: **magnetic fields** produced by an **alternating** current is a changing **magnetic field** (Its direction and magnitude changes with time

DC the **field** is constant both in magnitude and direction.

Ref. http://dx.doi.org/10.1016/j.micromeso.2015.12.032

Applications – Magnetic Drug Delivery

Magnetically (DC/AC) triggered site specific drug delivery system using Fe_3O_4 as drug carrier

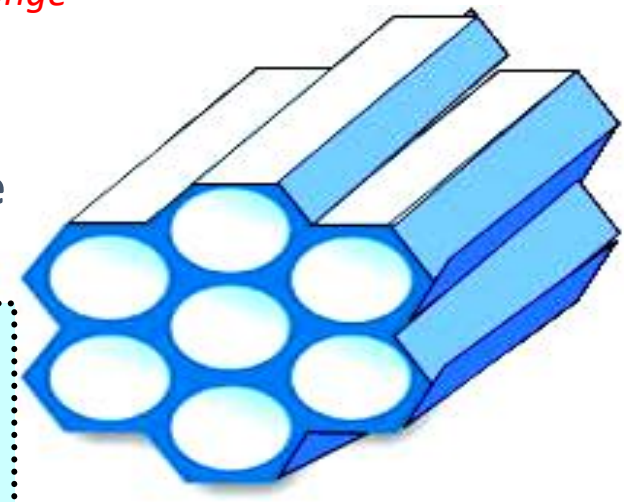
Importance of porous metals



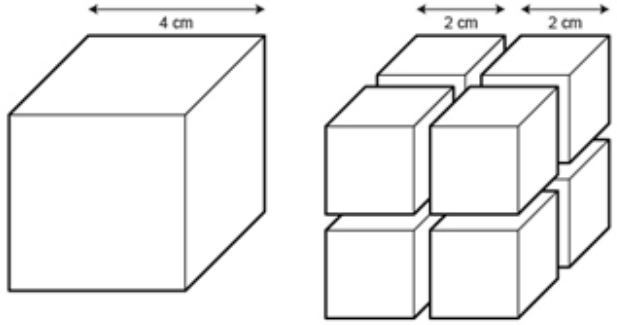


Sponge

Nanoporous structure



Advantage:
(i) Increase of surface area
(ii) Accessible mesopores



Surface area
= (4 cm x 4 cm x 6 faces) = 96 cm^2

Surface area of one cube
= (2 cm x 2 cm) x 6 faces = 24 cm^2

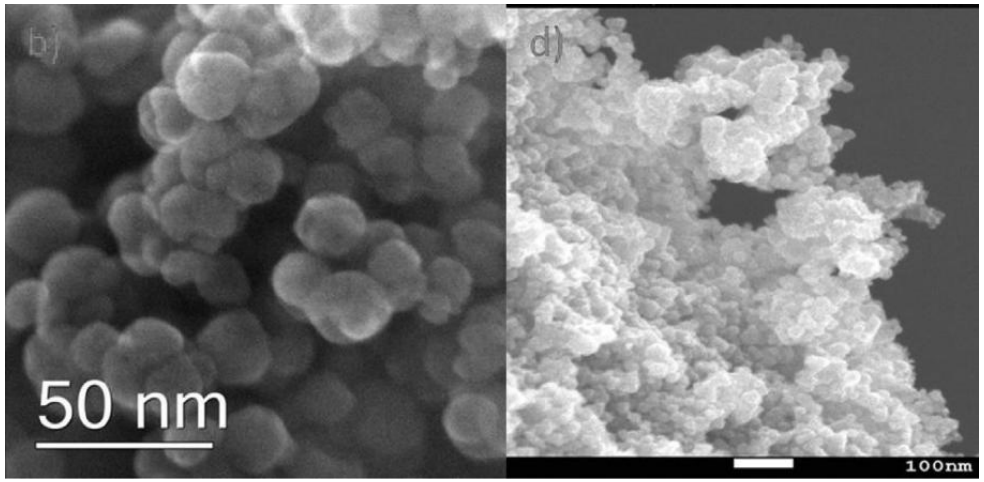
Total surface area
= 24 cm^2 x 8 cubes = 192 cm^2

Ref. http://dx.doi.org/10.1016/j.micromeso.2015.12.032

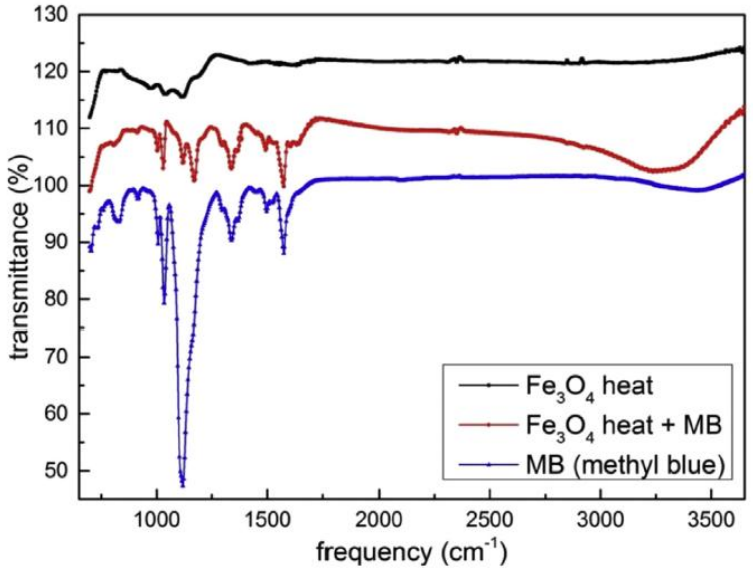
http://www.bbc.co.uk/schools/gcsebitesize/science/21c/materials_choices/nanotechnologyre shtml

Applications – Magnetic Drug Delivery

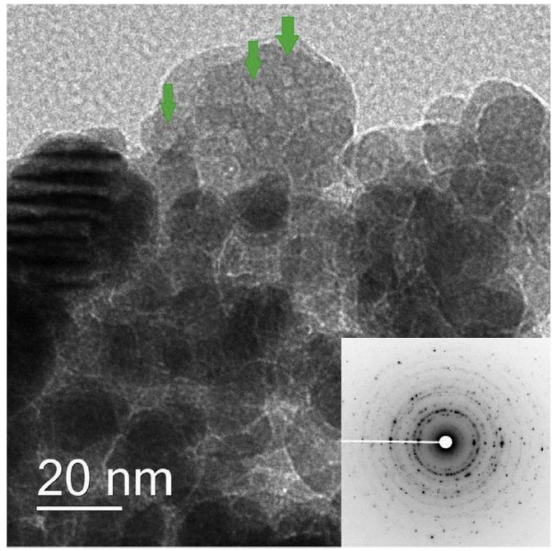
Magnetically (DC/AC) triggered site specific drug delivery system using Fe_3O_4 as drug carrier



Secondary electron (SE) images of Fe3O4 nanoparticles: STEM SE image of the surface of an agglomerate composed of 10-30 nm diameter particles; Detail of a) showing the smooth, rounded particles with extensive open porosity;



FTIR spectra of as-prepared porous Fe3O4, porous Fe3O4 after exposure to methyl blue, and methyl blue.



Bright field TEM image showing crystalline particles 10-50 nm in diameter. Extensive 2-5 nm diameter porosity is evident within the larger particles (green arrows)

Porous Fe3O4 structures have been used as a vehicle for the transport of drugs (Methylene blue as model drug)

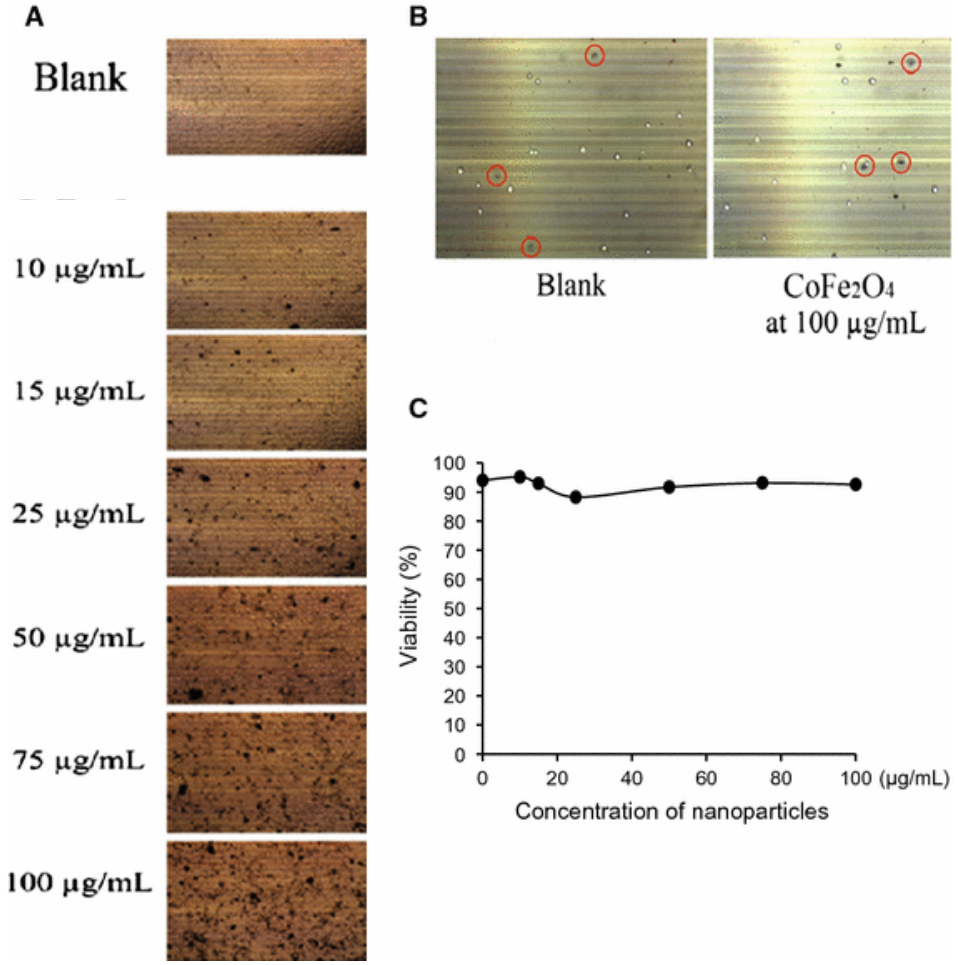
1. Crystalline Fe3O4 nanoparticle
2. Pores form from individual crystallite and between crystallites
3. Metheylene Blue as model drug
4. Adsorbed onto the porous surface through electrostatic interaction.

Ref. http://dx.doi.org/10.1016/j.micromeso.2015.12.032

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Cytotoxicity Test of the Fe_3O_4 nanoparticles

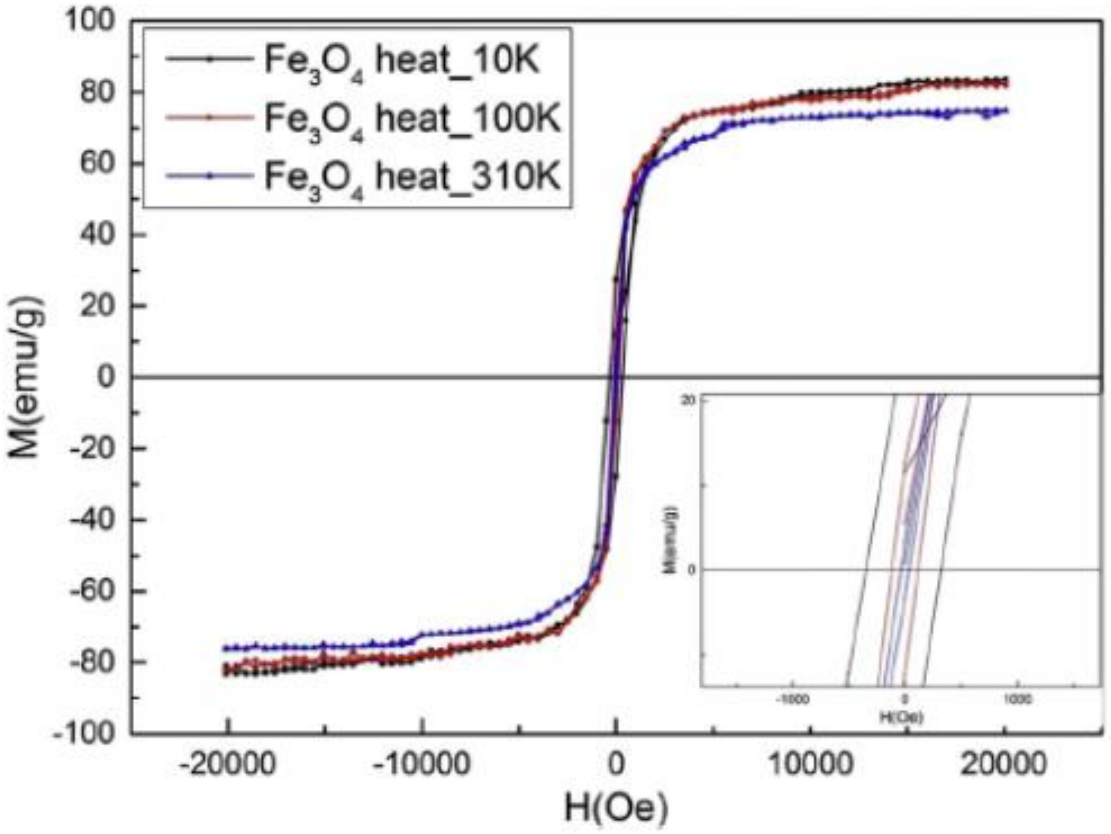


1. Cultures of A549 human cancer cells
2. Different concentrations of nano-conjugates ranging from 10 to 100 μg/ml
3. Cell viability is more than 90%

Ref. http://dx.doi.org/10.1016/j.micromeso.2015.12.032

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The magnetic hysteresis loops of porous Fe3O4 at temperatures of 10, 100, and 310 K. The zoomed low-field region (inset) shows values of the coercive field.

Ref. http://dx.doi.org/10.1016/j.micromeso.2015.12.032

Magnetic properties of the Fe3O4 nanoparticles

The S-shaped hysteresis loops with negligible coercive field (HC) are a typical feature of superparamagnetic Nanoparticles

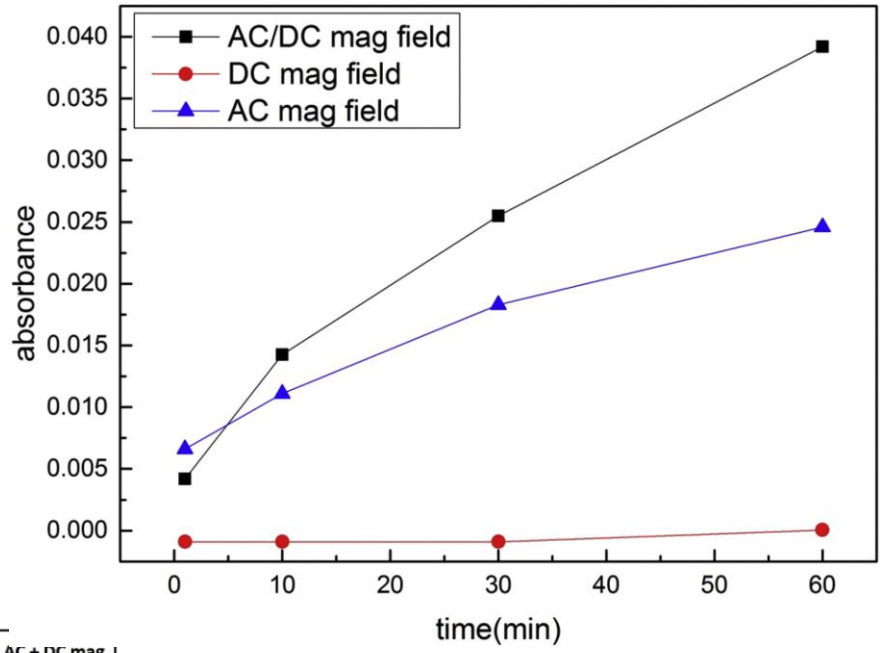
The saturation moment (Ms) measured at 10K is quite high and has a value of 85.5 emu/gm, while at 310 K the saturation moment was slightly smaller at around 77 emu/gm.

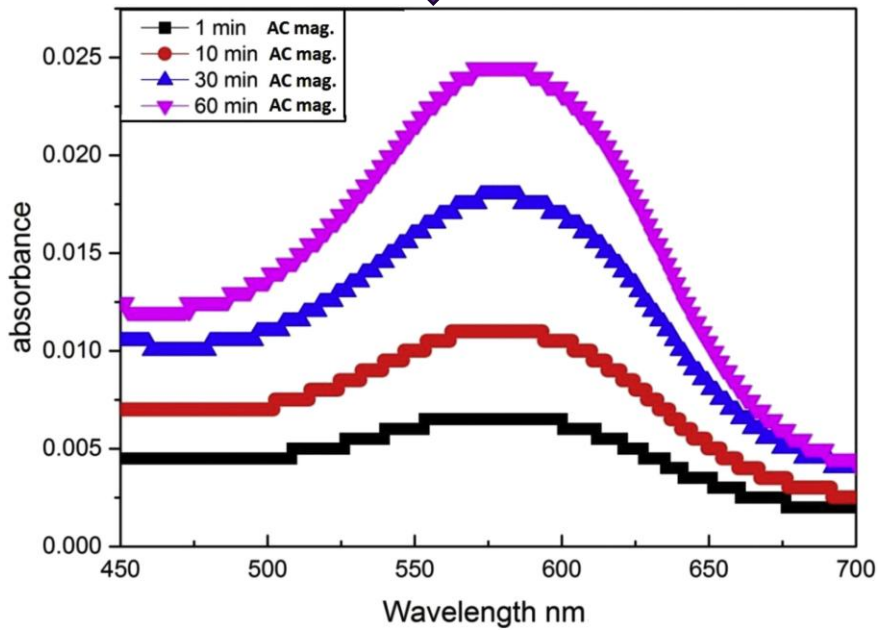
Small Hc values were obtained for the nanoparticulate material, because the magnetic moment of each nanocrystallite making up this material was small enough to become unsettled by thermal excitation

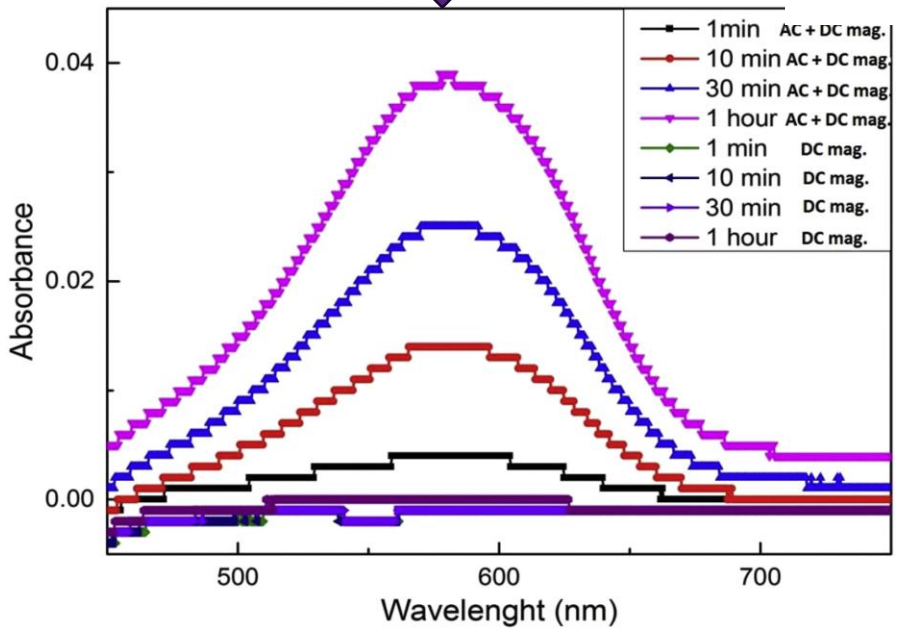
Magneto-Mechanical Nano-machineries for Precision Drug Release









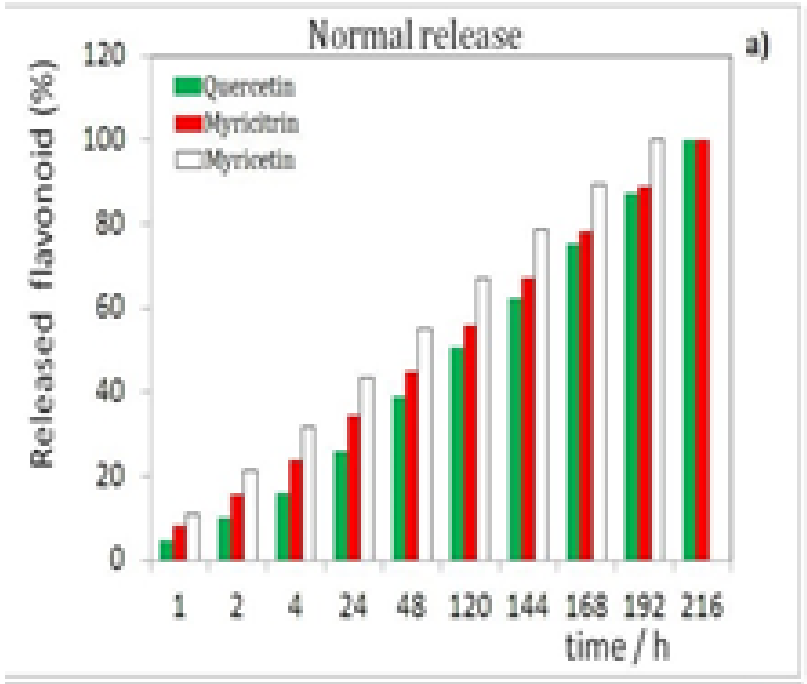


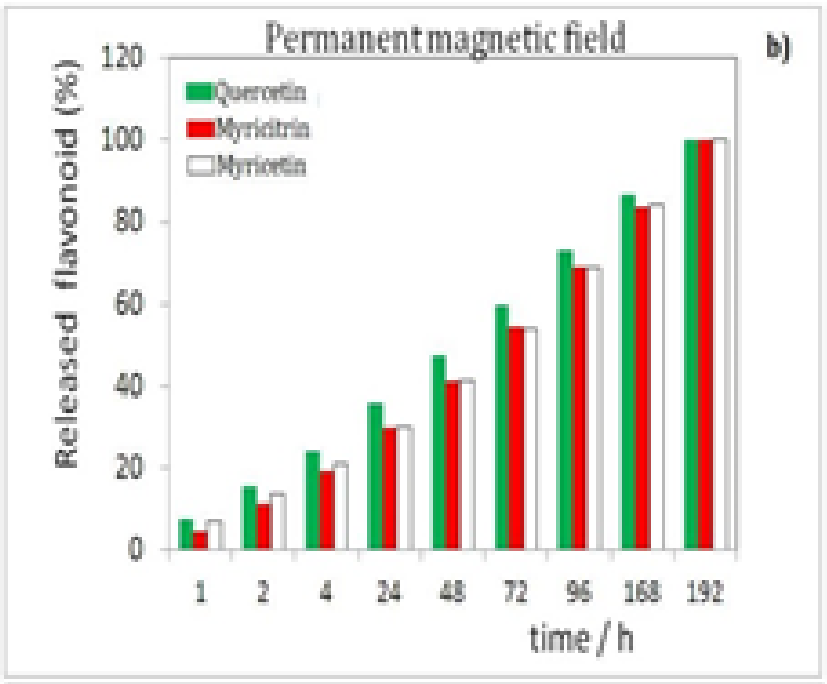
The released model drug has a strong absorption band around **580 nm**. This wavelength acts like the drug's **optical fingerprint**.

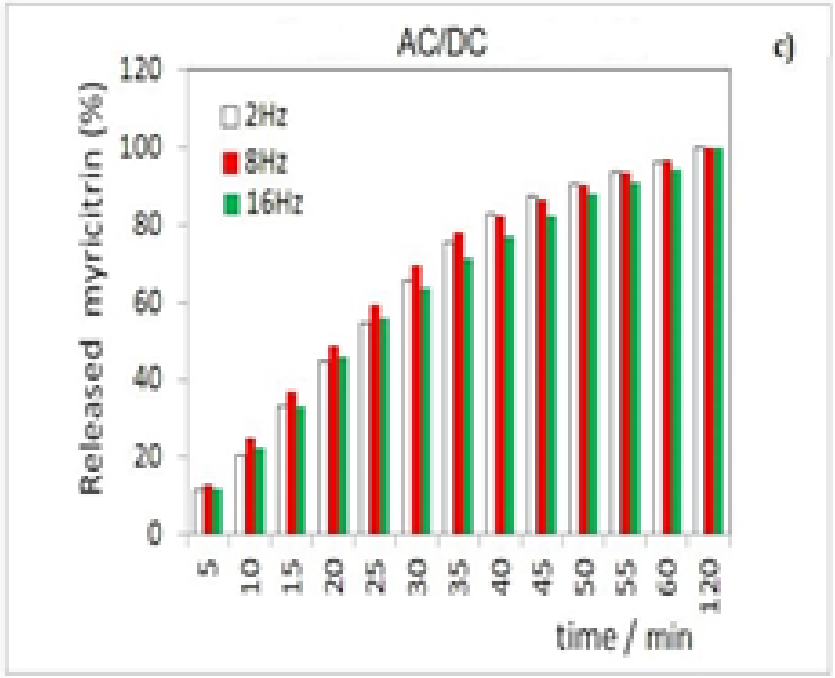
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Same mechanism using real drugs:







Cumulative released profiles of quercetin, myricetin and myricitrin in (a) normal condition, (b) permanent magnetic field and (c) and AC/DC magnetic fields, 2Hz, 8 Hz and 16 Hz.

100% drug can be released within 2 hrs if combined AC/DC technique is used where 100 times faster compared to normal release and only DC field is used.

Ref. http://dx.doi.org/10.1016/j.micromeso.2015.12.032

Practical contribution of Brownian and Neel relaxation in site-specific delivery

1.Guidance with external magnetic fields

1. When nanoparticles are circulating in blood, both Brownian and Néel relaxation matter.
2. Brownian relaxation affects how quickly particles reorient with the guiding field → helps efficient steering.
3. Néel relaxation ensures fast internal magnetic moment flips → maintains strong response even if Brownian rotation is slowed by viscous environments (like blood plasma).

2.Hyperthermia for therapy

1. In magnetic hyperthermia (using AC fields to heat particles at the tumour site), energy dissipation comes from relaxation.
2. Small particles (Néel-dominated) flip internally to generate heat.
3. Larger or coated particles (Brownian-dominated) rotate physically in fluid to dissipate heat.
4. Which dominates depends on the core size and hydrodynamic size.

3.Drug release mechanism

1. If nanoparticles are designed with a drug-loaded coating, Brownian rotation can induce *mechanical agitation* to help drug release.
2. If particles are immobilised at the target site, only Néel relaxation can continue to dissipate energy → useful for triggering local heating and drug release without particle rotation.

AC/DC magnetic syringe device for stimulated drug release, injection and ejection of nanocarriers

The University Of Queensland AUSTRALIA

What factors need to be considered here to optimise for successful magneto-mechanical site-specific drug release using Fe_3O_4?

Drug loading into carrier:

1. Porosity
2. Size
3. High surface to volume ratio
4. Electrostatic interaction
5. Adsorption
6. Self vibration
4. Collision between the particles
5. Collision between particles and media.

Challenges:

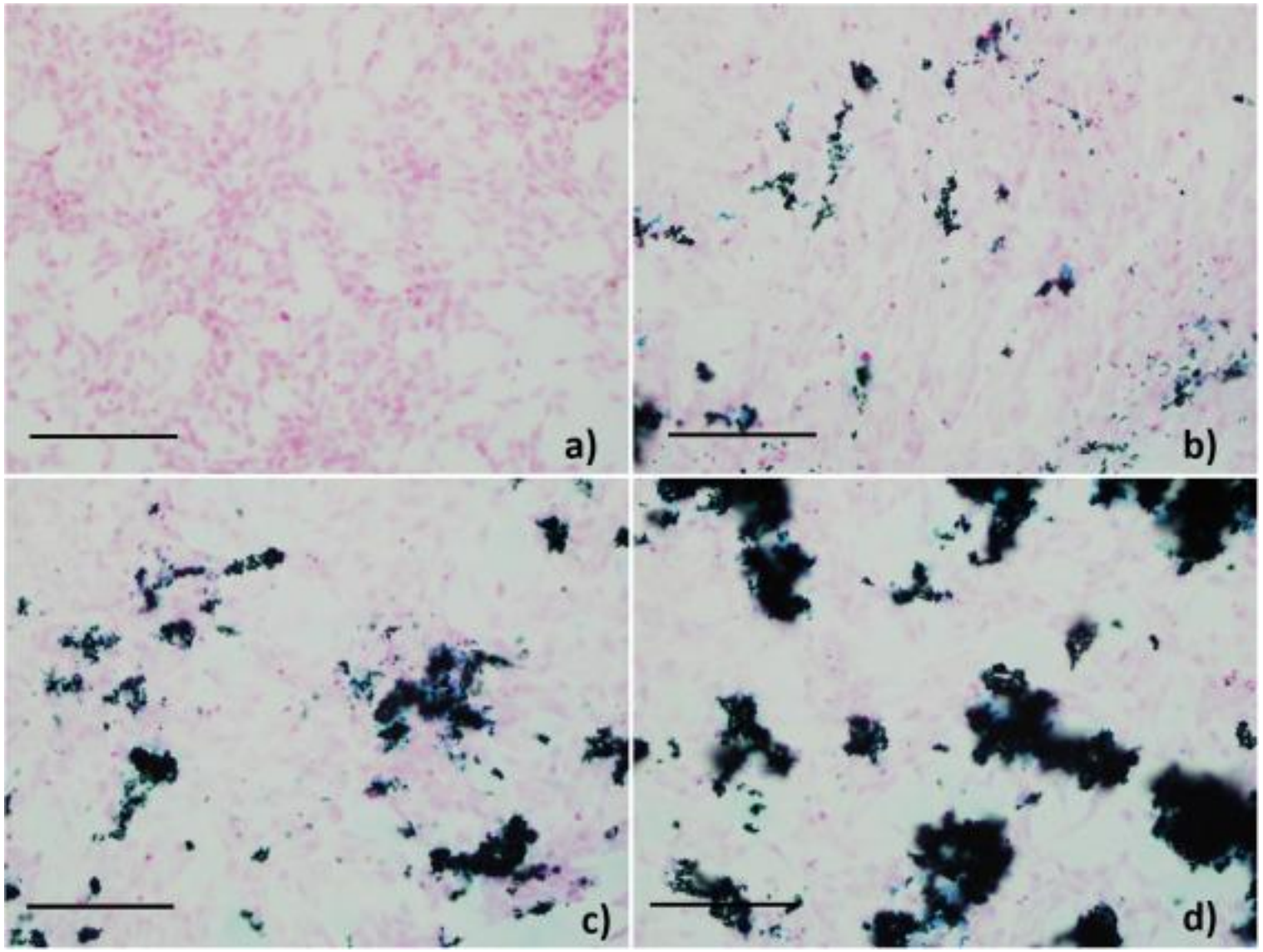
Removing the nano-carrier from body and re-use?

Site specific drug release:

1. Cytotoxicity
2. DC magnetic field
3. AC magnetic field
4. Light absorption (release evidence
5. magnetic moment of the particle.
6. Frequency
7. Amplitude
8. Applied current
9. DC Magnetic field
10. AC magnetic field
11. Coil design
12. Viscosity of the media

https://www.sciencedirect.com/science/article/pii/S0304389420329095
https://www.sciencedirect.com/science/article/pii/S0304885319310236

Magnetic drug delivery based on superparamagnetic porous nanostructures

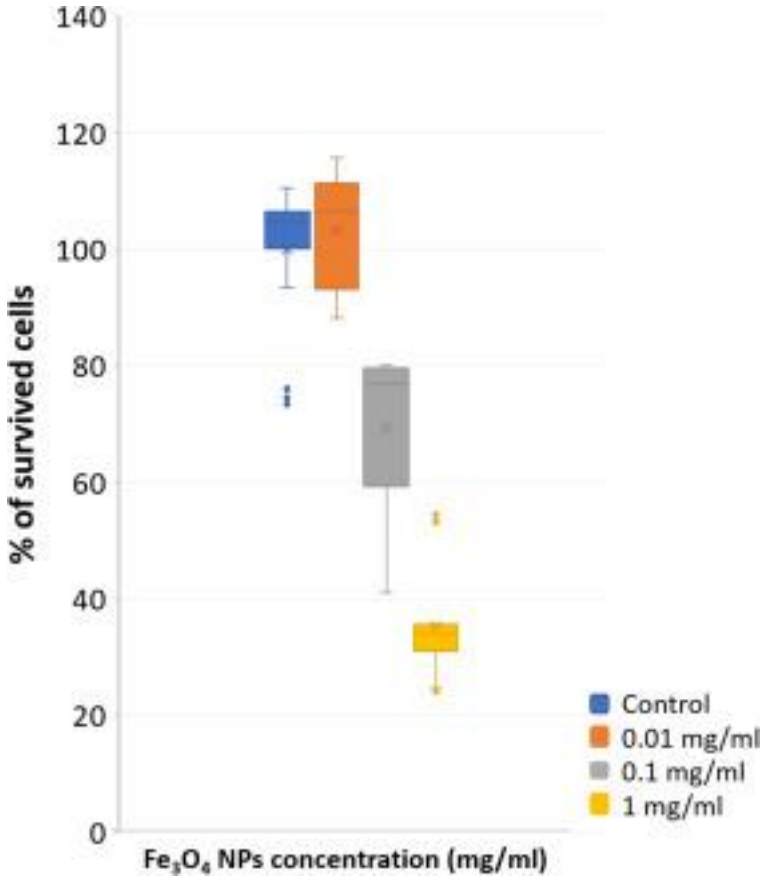


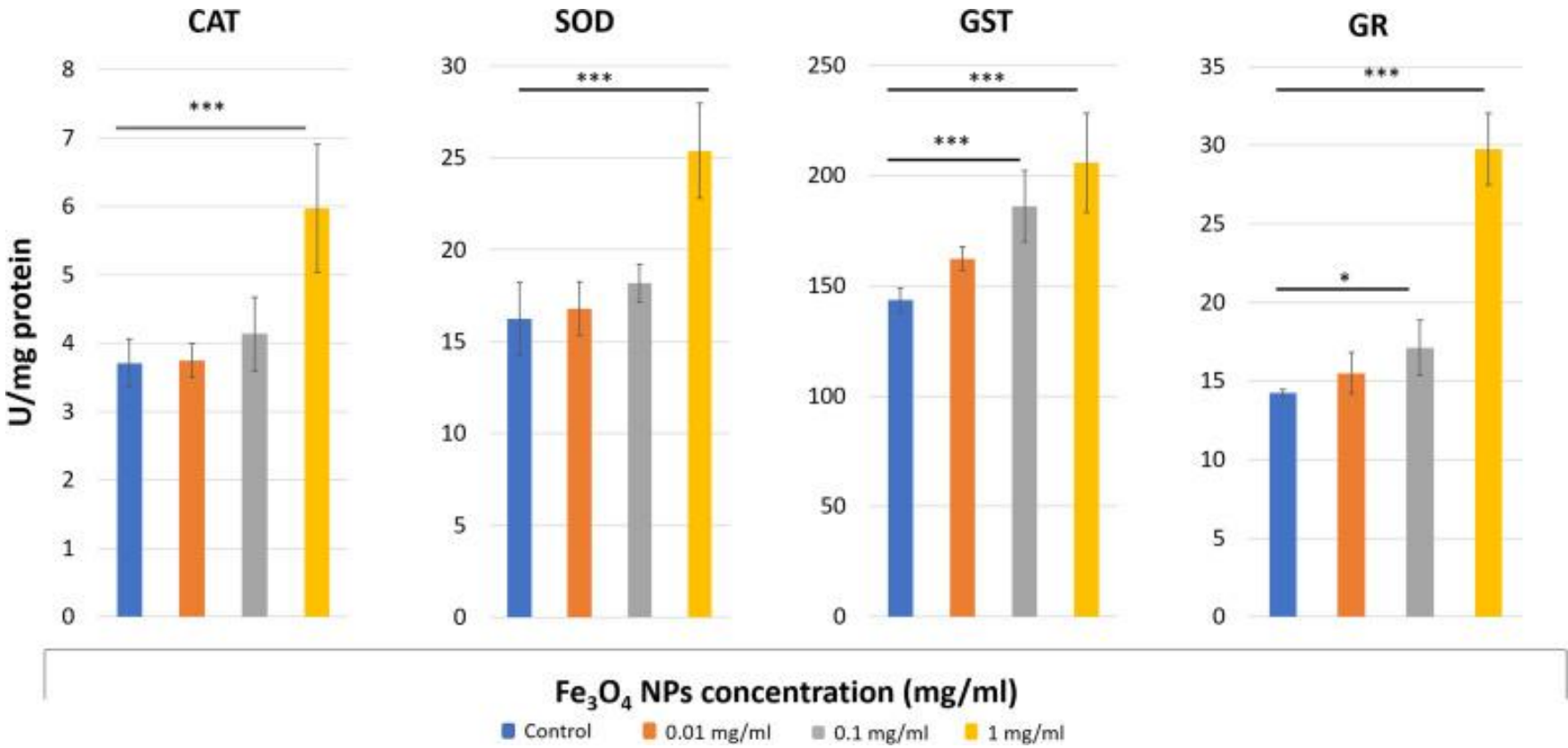
Prussian blue staining against iron on differentiated SH-SY5Y human neuroblastoma cell line at 24 h post exposure to different concentrations of Fe_3O_4 nanoparticles; A – 0 mg/ml (control), B – 0.01 mg/ml, C – 0.1 mg/ml, D – 1 mg/ml; Blue – metabolised iron positive stain, Red – nuclei and cytoplasm, Black – non metabolised nanoparticles; magnification 400 ×; Scale 100 μm. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article).

J. of Hazardous Materials 2021, 409, 124918 -https://www.sciencedirect.com/science/article/pii/S0304389420329095#fig0055

CRICOS code 00025B

Higher concentration of particles (Fe3O4) inside biological cell is toxic



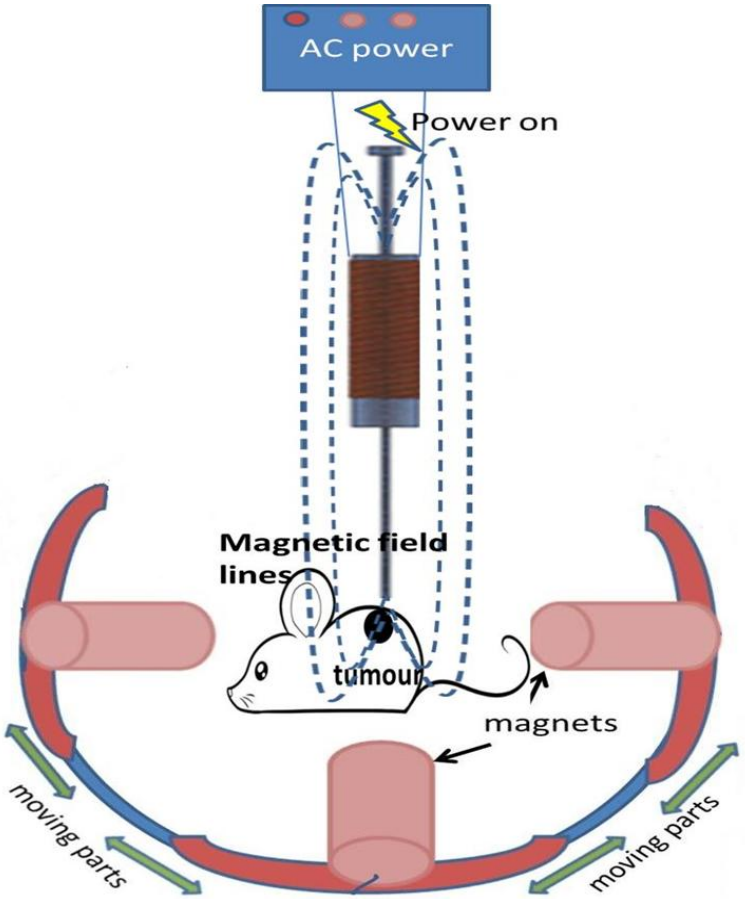


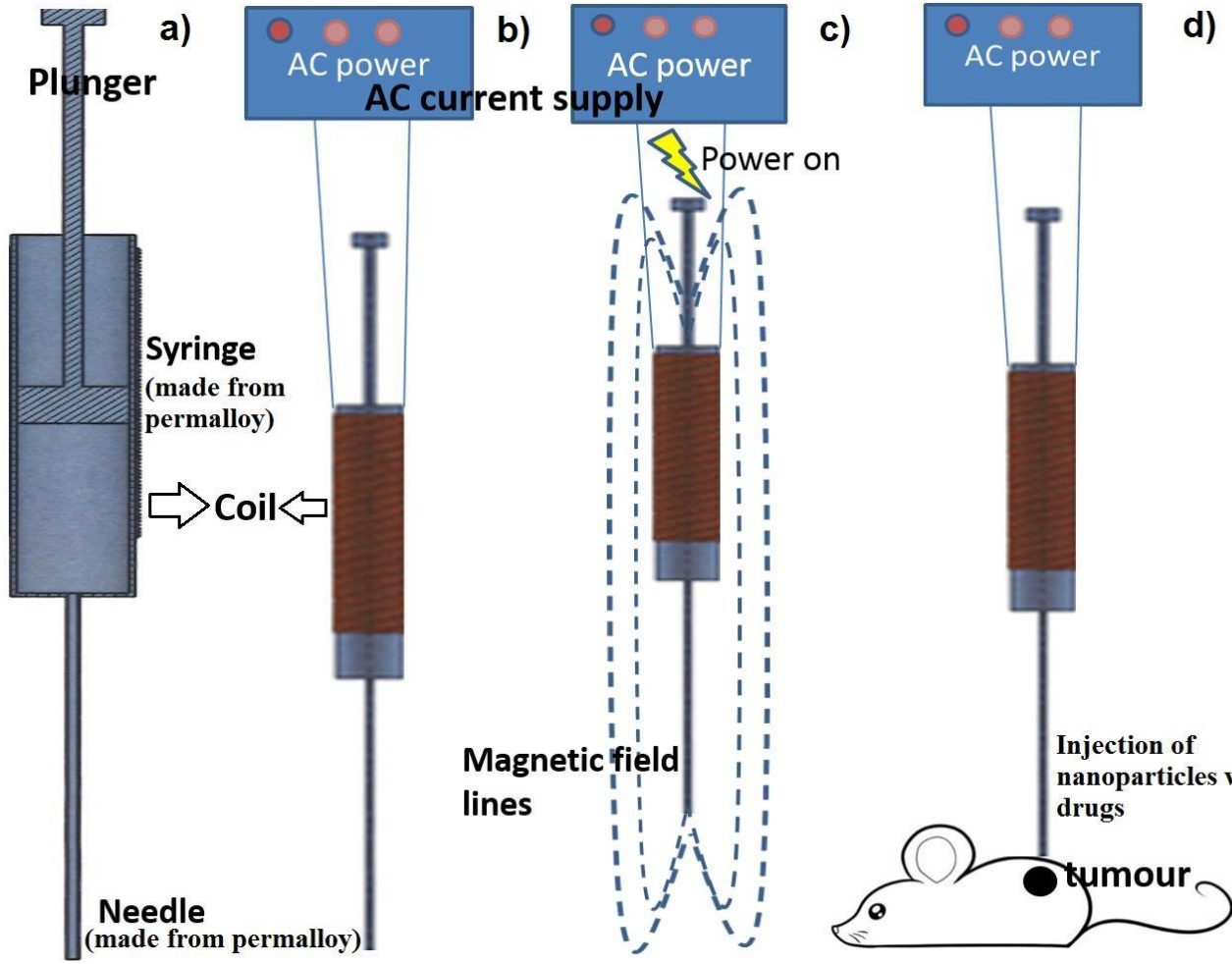
J. of Hazardous Materials 2021, 409, 124918 -https://www.sciencedirect.com/science/article/pii/S0304389420329095#fig0055

Applications – Magnetic Syringe

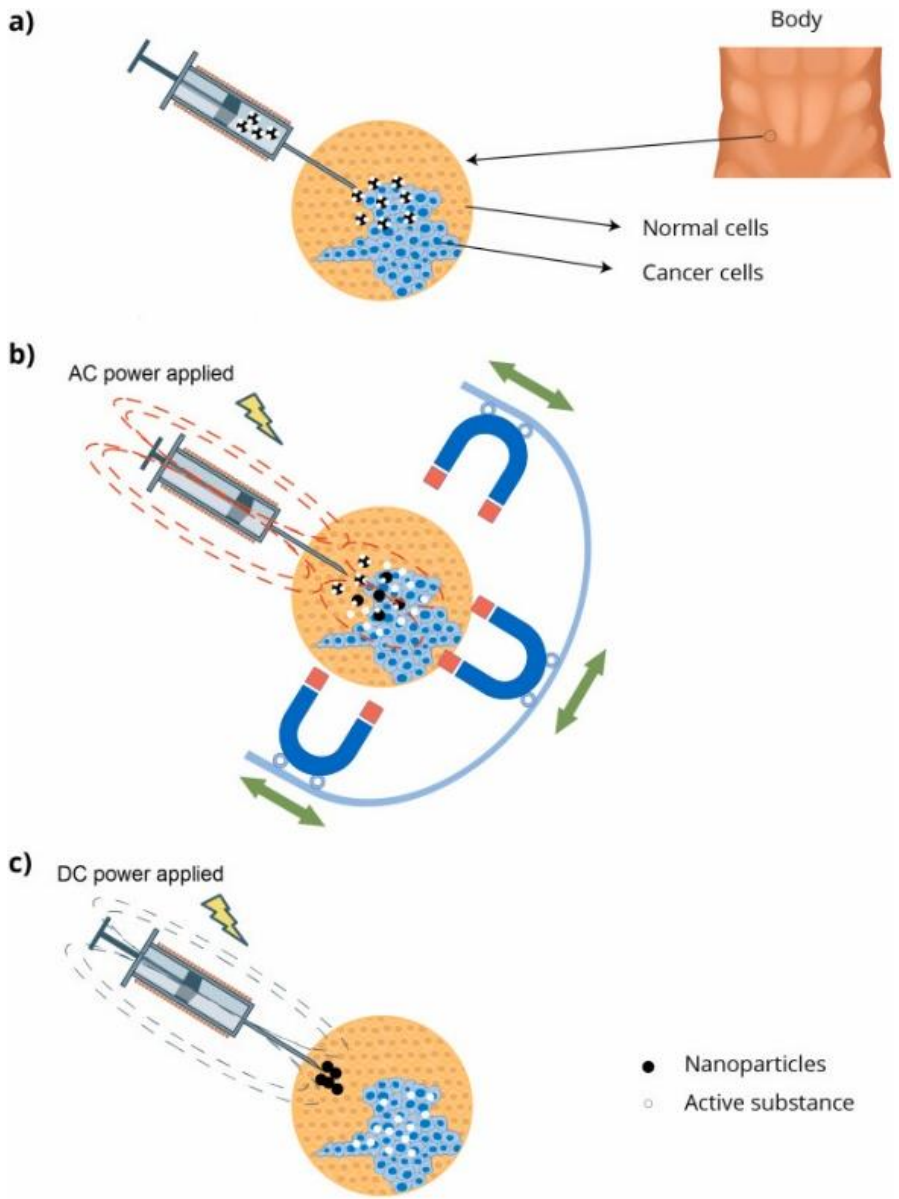
THE UNIVERSITY OF QUEENSLAND AUSTRALIA

Magnetically (DC/AC) triggered site specific drug delivery system using Fe_3O_4 as drug carrier

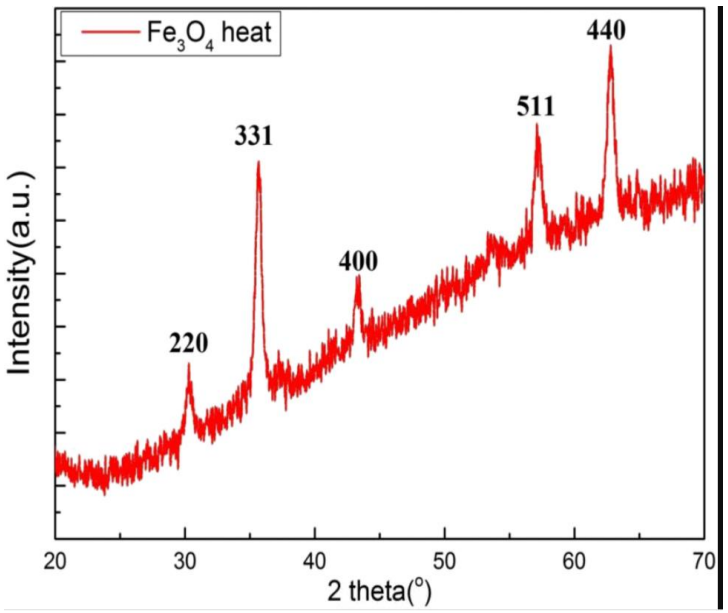


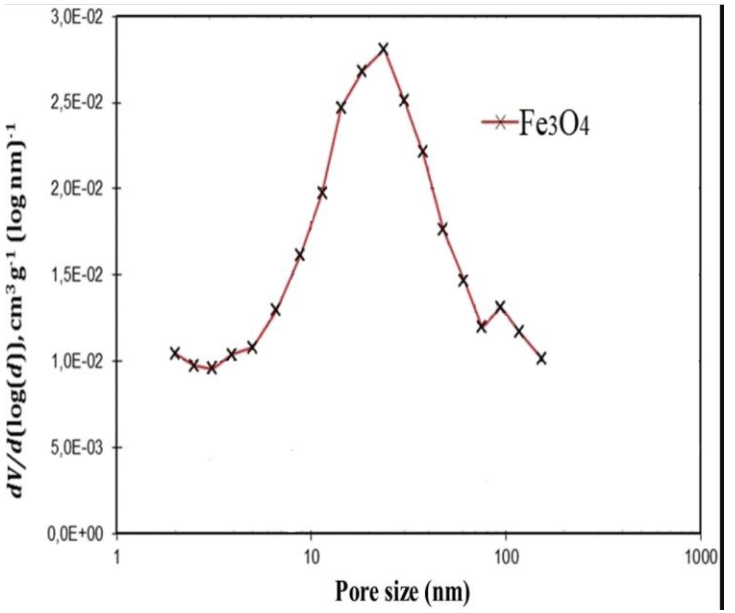


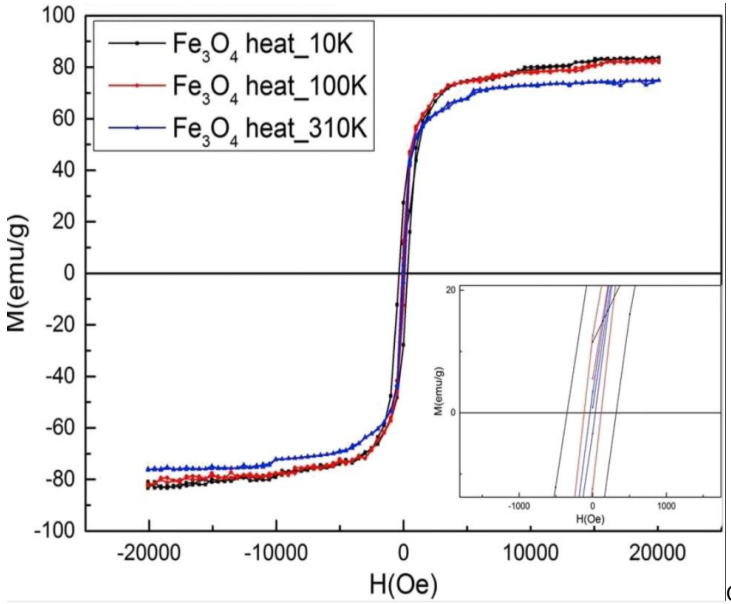
Magnetic syringe

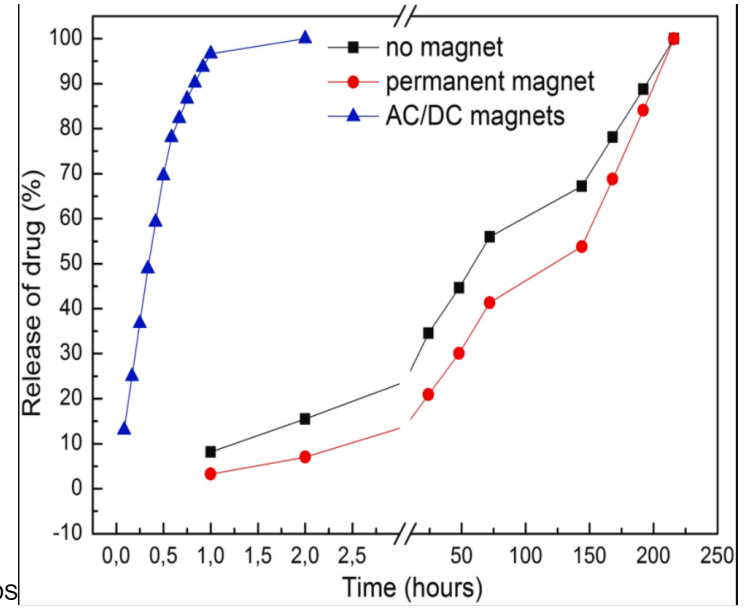


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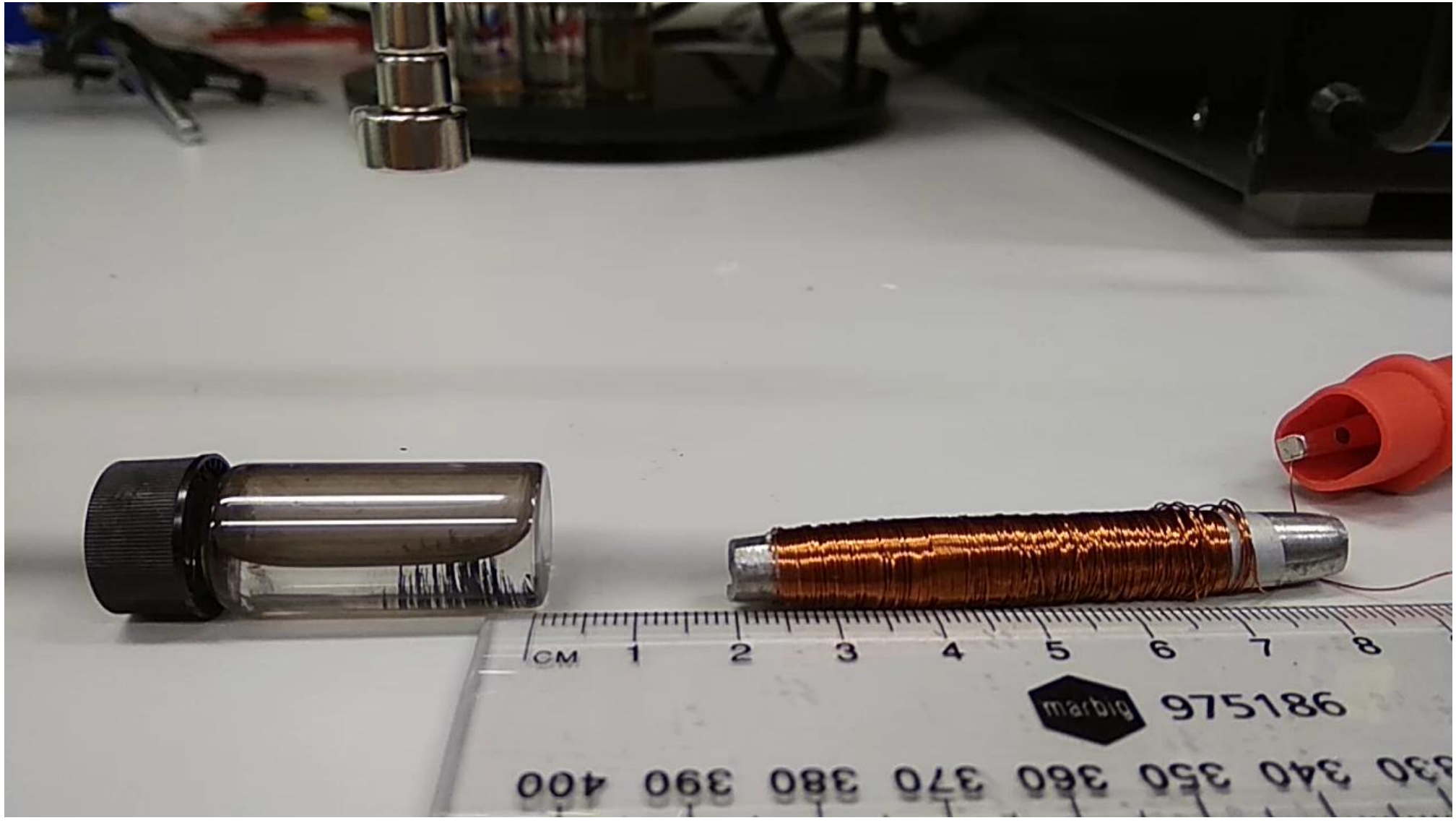


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Applications – Magnetic Syringe

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Magnetically (DC/AC) triggered site specific drug delivery system using Fe_3O_4 as drug carrier



Lab Practical on Electrochemical Characterisation of Magnetic Nanoparticles: